



Physics-informed and Data-driven Methods for Developing Detective and **Preventive** Cyber-physical Defenses

Security-aware Data Sharing

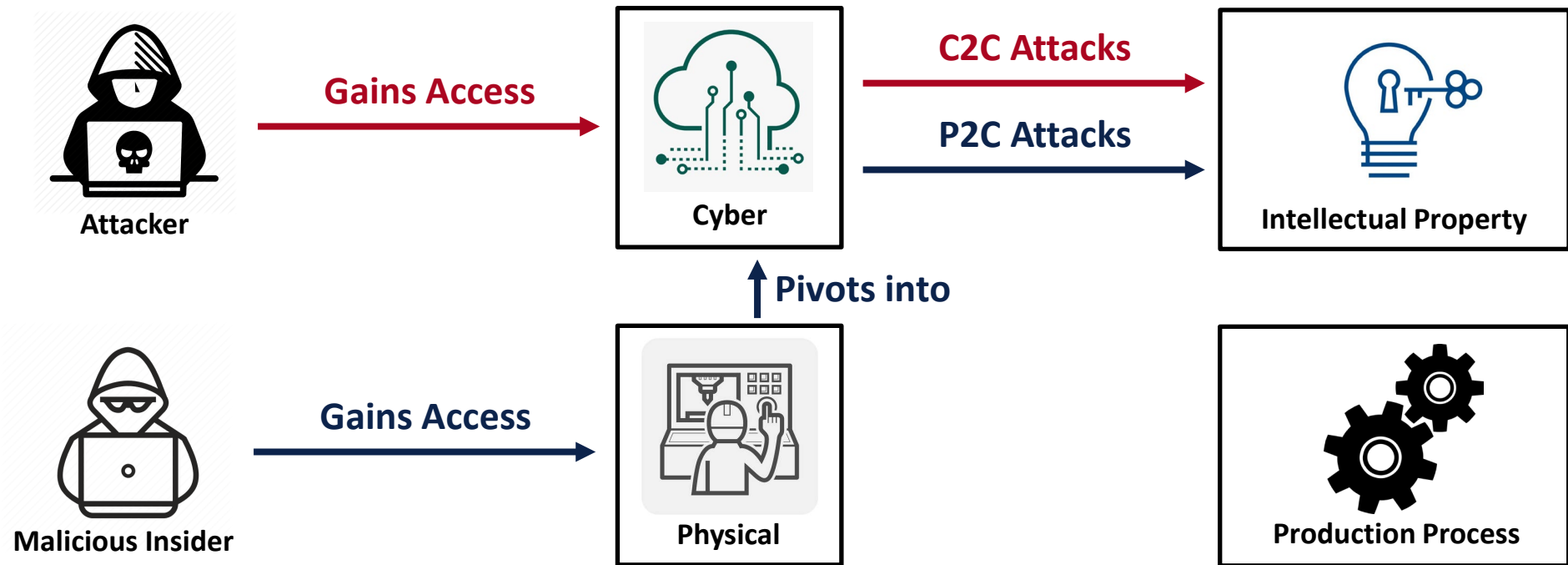
Habibor Rahman

Assistant Professor, Mechanical Engineering
University of Massachusetts Dartmouth



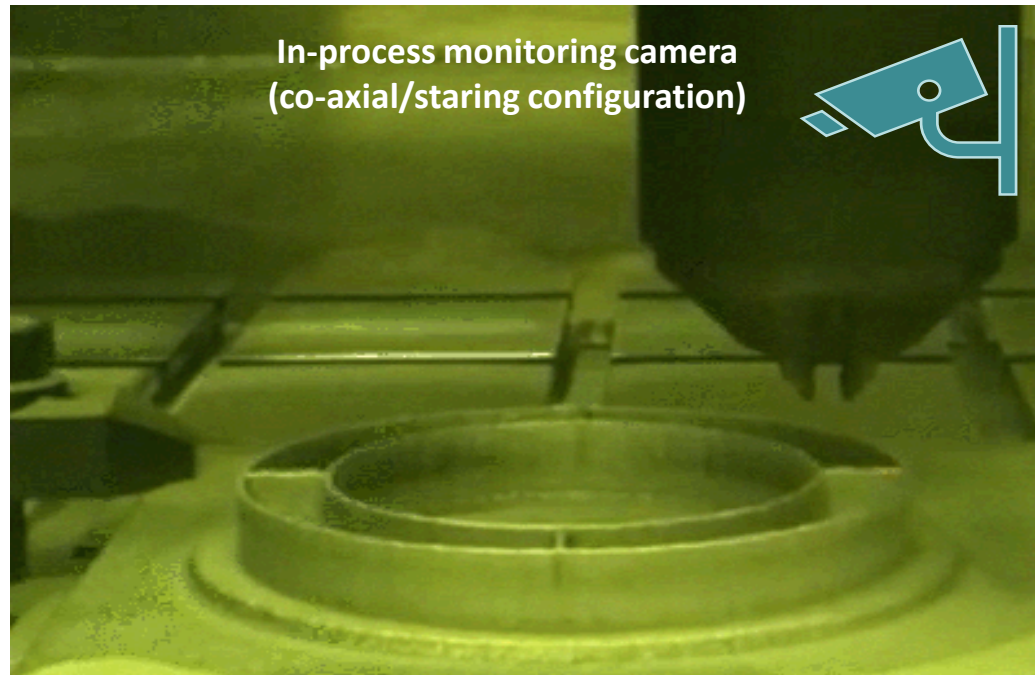
Motivation – C2C and P2C Attacks

- C2C attack example: Adversaries stealing process monitoring data to discover design intent and process information
- P2C attack example: Insiders/collaborators sharing confidential data

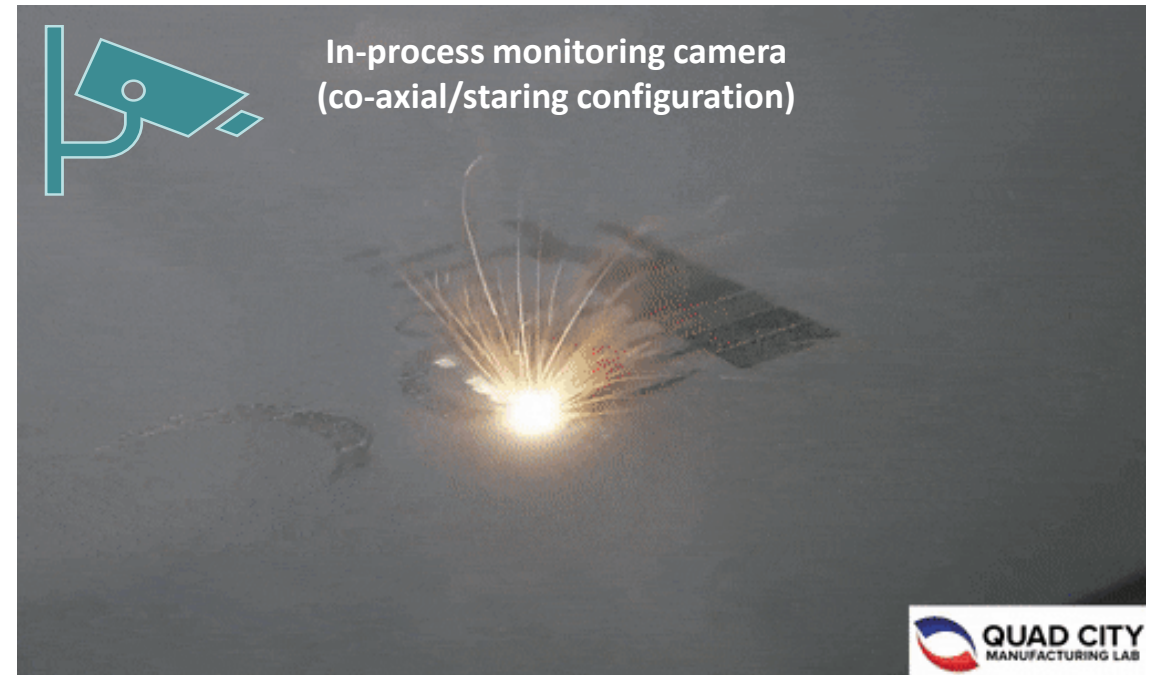


Background – Metal AM Processes

- **Metal Additive Manufacturing (AM) processes** fabricate parts based on a digital 3D model by selectively melting metal wires and/or fine metal powders layer by layer.



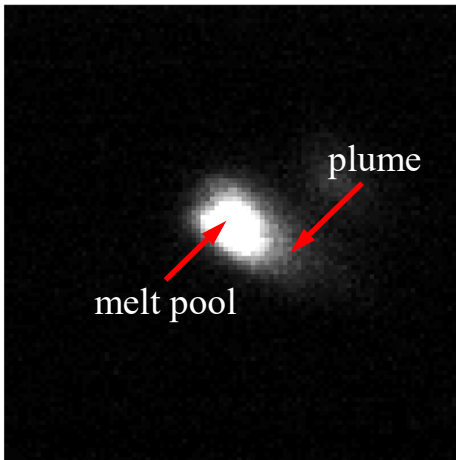
Directed-Energy Deposition (DED) process
Source: [NASA/JPL](#)



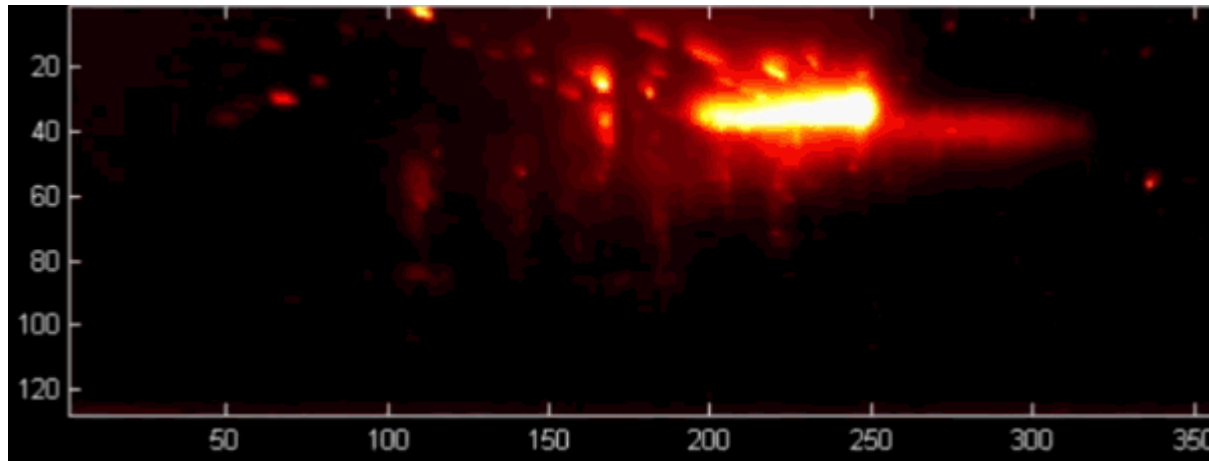
Laser Powder Bed Fusion (LPBF) process
Source: [Gfycat](#)

Background – Data Privacy in Metal AM

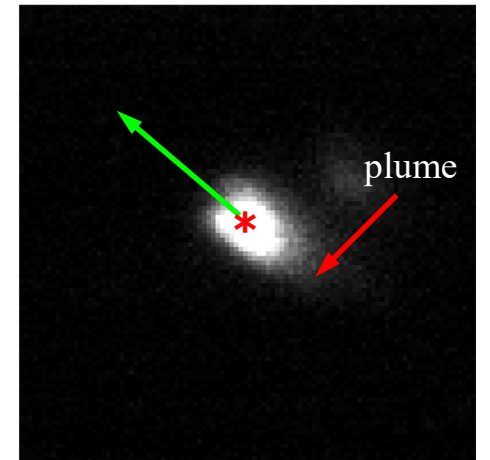
- **Melt pool images** collected by the co-axial camera are the most widely used in-process monitoring data in metal AM process (ASTM E3353-22).
- The plume in the melt pool image can reveal the direction of nozzle/laser movement, and thus the **printing/scanning path** information can be leaked.



Typical meltpool image



Laser is melting metal powders

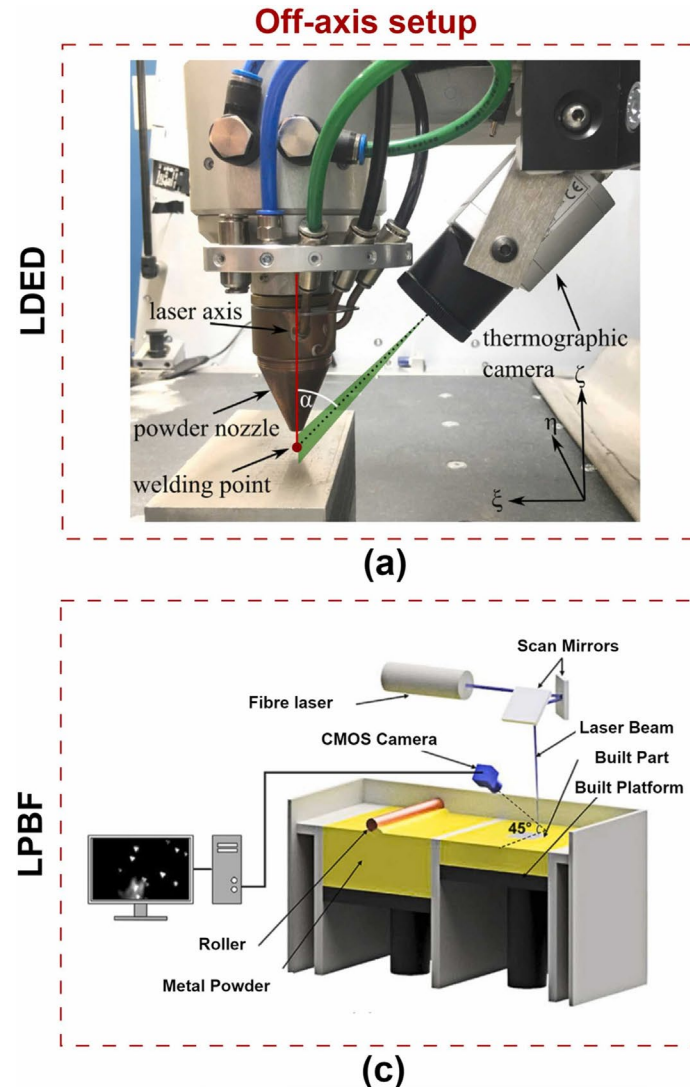


Printing/scanning
direction

Background – Data Collection

Off-axis setup

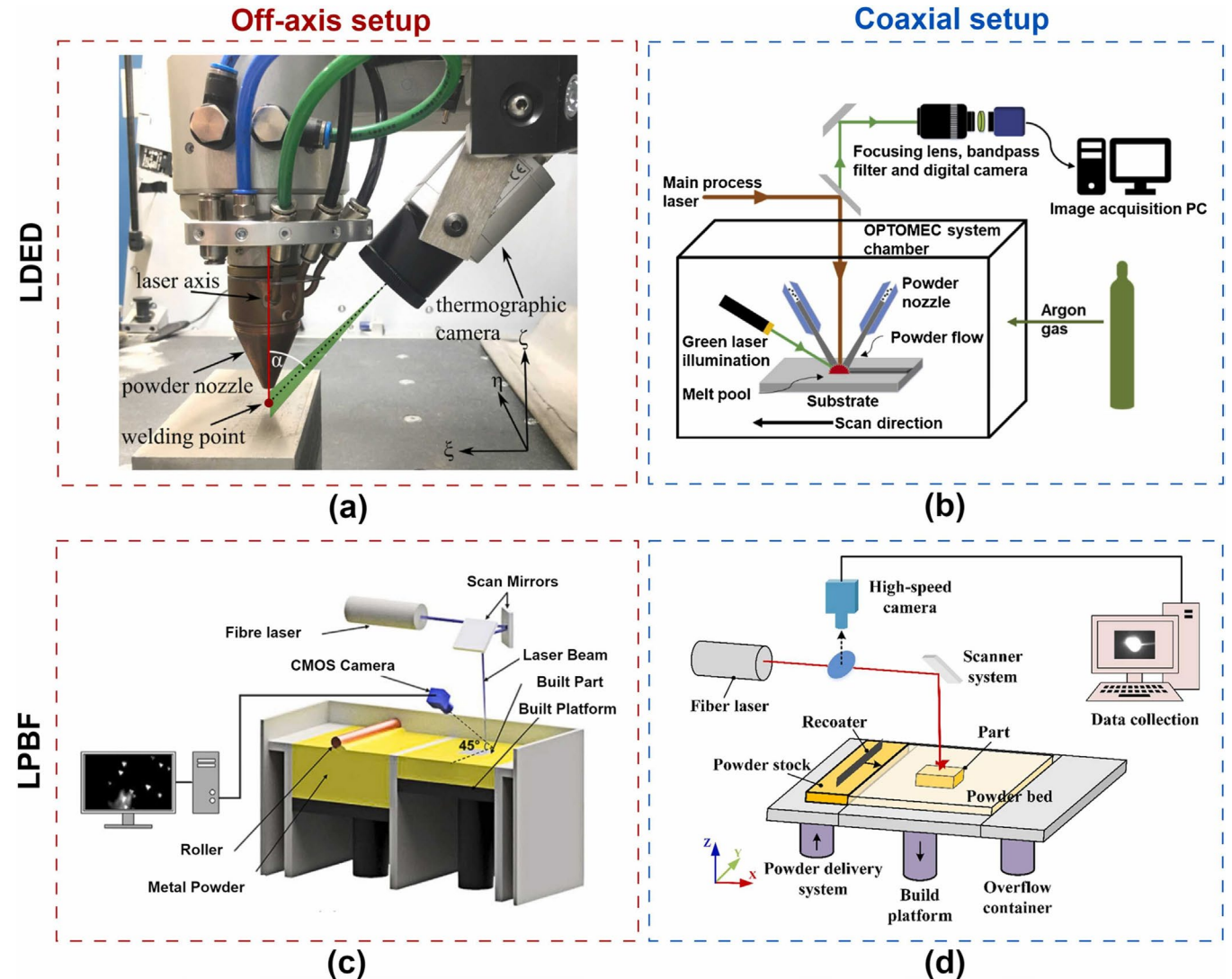
- Positioned at an angle to monitor the melt pool, powder bed, or part surface.
- Real-time visualization of process features (e.g., surface morphology, thermal gradients).
- **Advantages:** Broader field of view; less obstruction by processing equipment.
- **Limitations:** Potential for distortion or occlusion due to viewing angle.



Background – Data Collection

Coaxial setup

- Aligned with the laser or energy source, capturing data directly along the processing axis.
- Precise monitoring of the melt pool dynamics, energy distribution, and material deposition.
- **Advantages:** High spatial resolution and direct process insight.
- **Limitations:** Limited field of view; higher integration complexity.



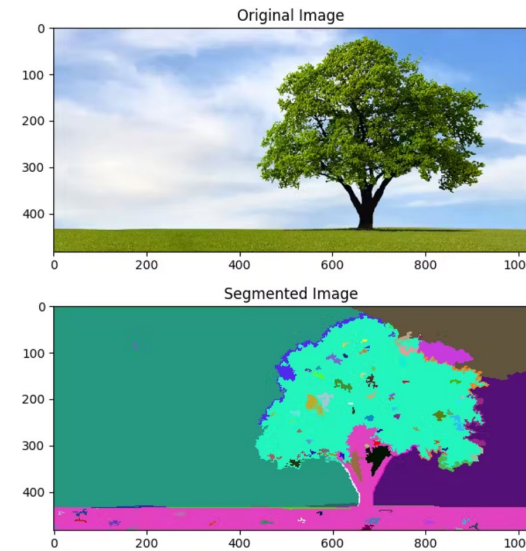
Background – Data Processing

Preprocessing

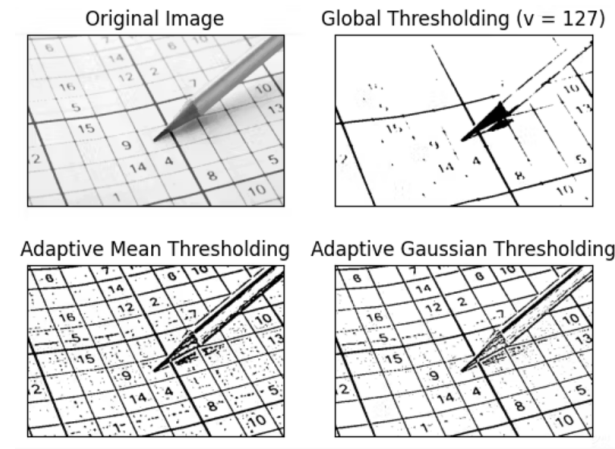
- **Noise Reduction**: Application of filters (e.g., Gaussian, median) to remove sensor noise
- **Contrast Enhancement**: Techniques like histogram equalization to improve the visibility of key features.
- **Alignment and Registration**: Synchronization of multi-view or multi-sensor data for coherent analysis.

Segmentation

- **Region-Based Segmentation**: Isolating specific areas like the melt pool, solidified material, or defects.
- **Thresholding Techniques**: Binary or adaptive thresholding for distinguishing features like cracks or pores.
- **Machine Learning-Based**: Use of CNNs or U-Nets for automated, accurate segmentation.



Region-based segmentation

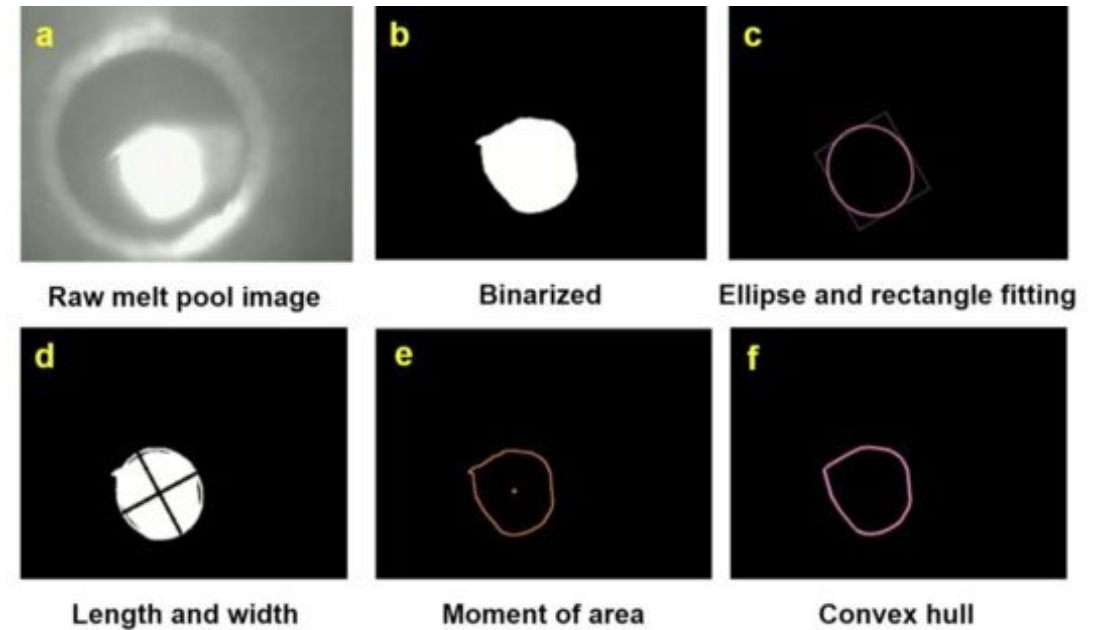


Thresholding technique

Background – Data Processing

Feature Extraction

- *Edge Detection*: Identify melt pool boundaries or part contours using Sobel, Canny, or similar algorithms.
- *Thermal Pattern Analysis*: Extraction of thermal gradients and hot spots from thermal imaging data.
- *Powder Bed Inspection*: Analysis of anomalies such as uneven powder distribution or contamination.

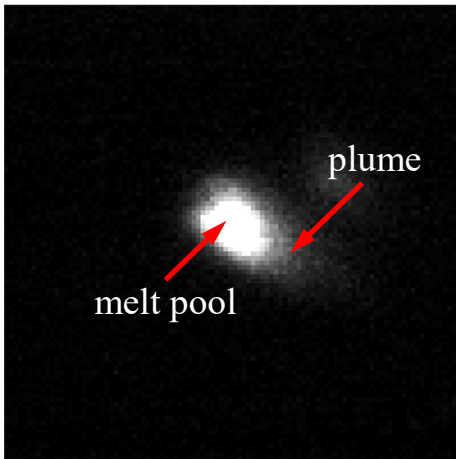


Coaxial melt pool image processing and
geometric feature extraction

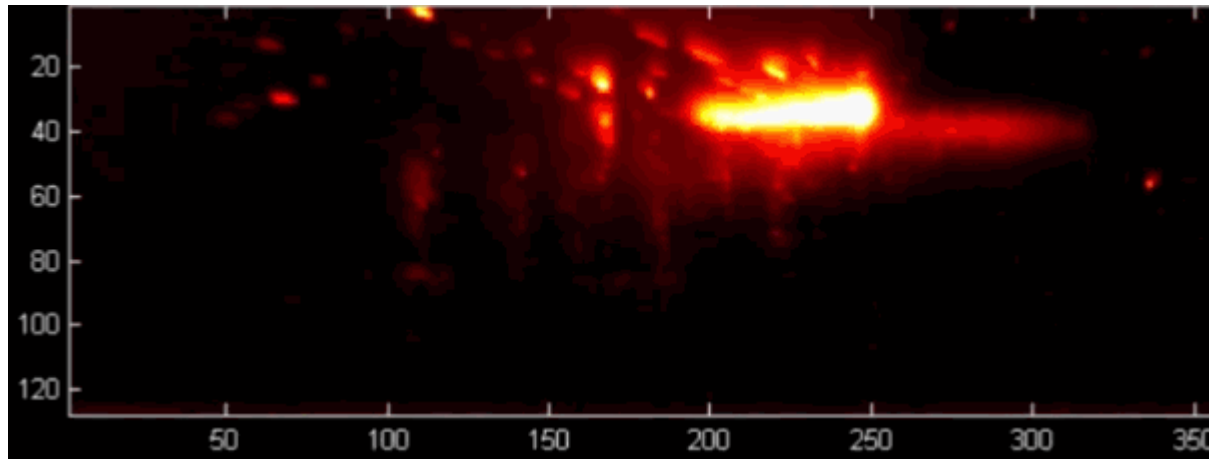
Source: <http://dx.doi.org/10.1016/j.rcim.2023.102581>

Background – Data Privacy in Metal AM

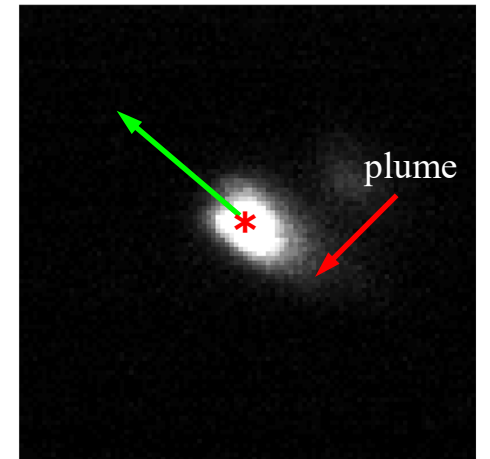
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Typical melt pool image



Laser is melting metal powders



Printing/scanning
direction

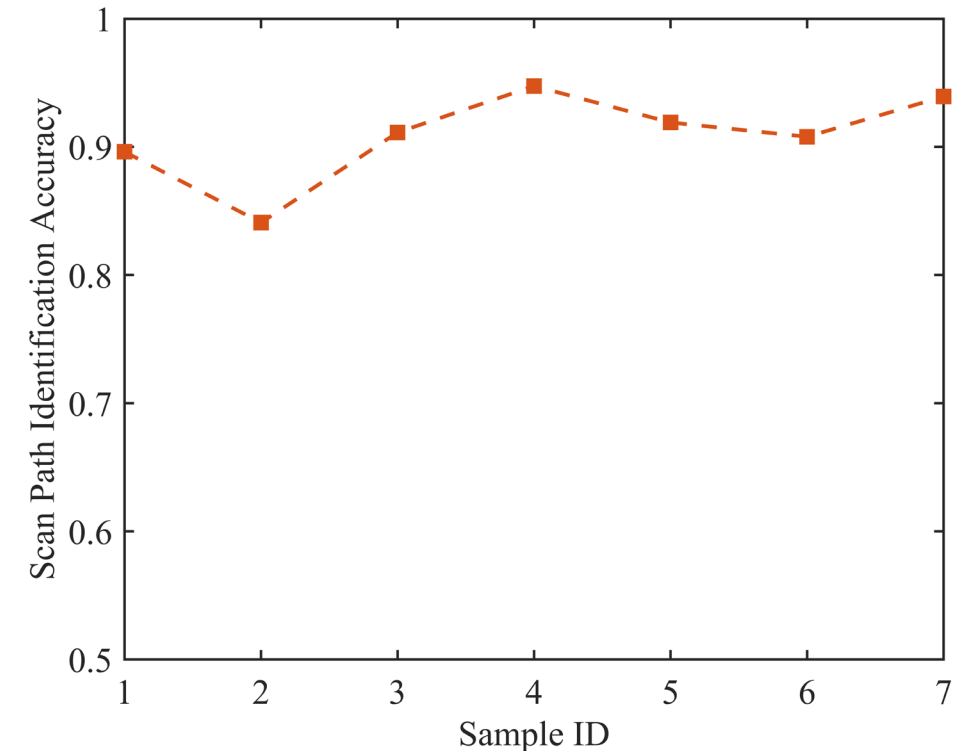
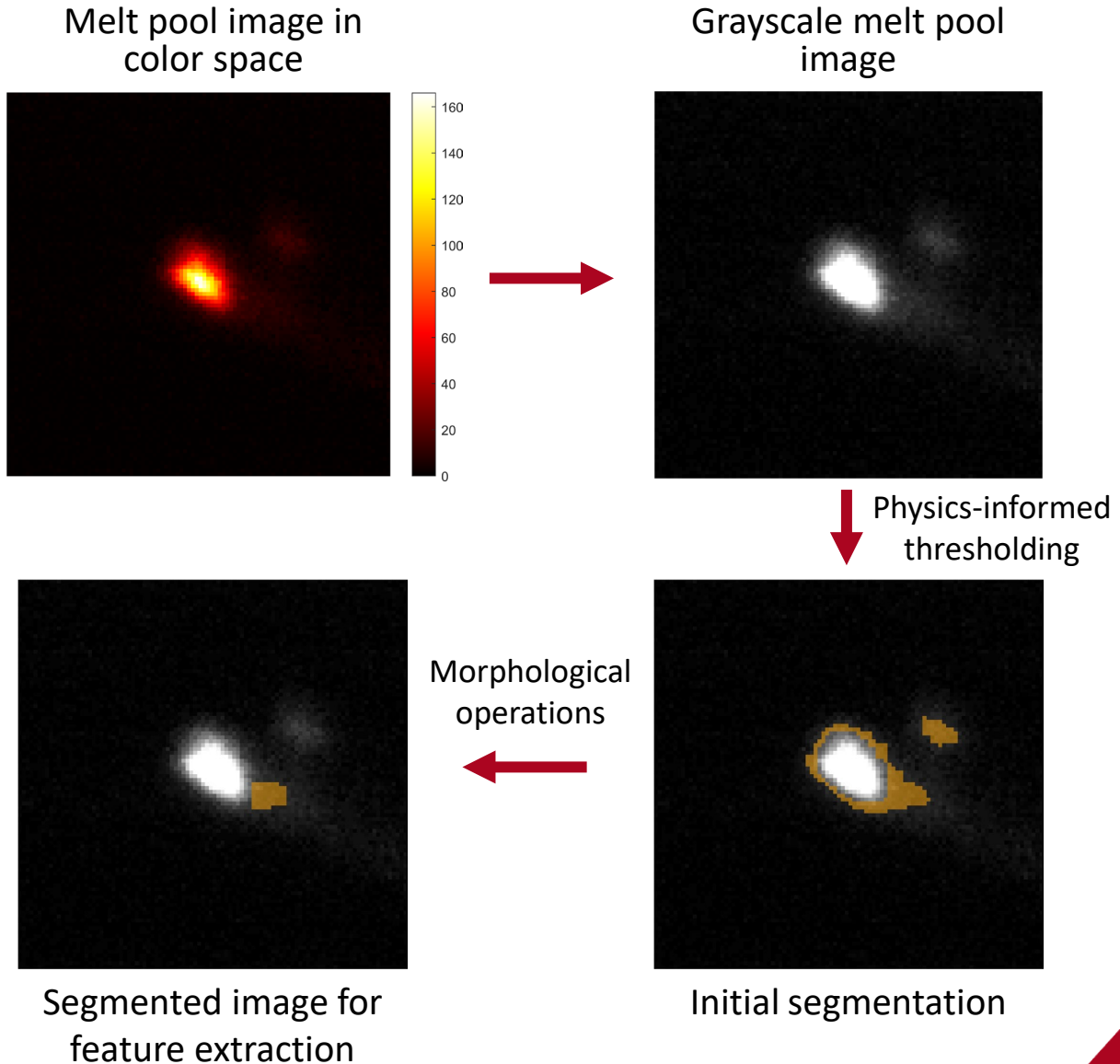
Scan Pattern Identification (SPI) Algorithm

Input: Set of **raw melt pool images** with coordinates $\{I_{raw_i}, \mathbf{c}_i\}_{i=1}^N$, image segmentation thresholds $\mathcal{T}_{\min}$, $\mathcal{T}_{\max}$, disk radius for morphological operations r_c, r_o

Output: Clusters of spatial coordinate points based on scan directions $(\mathbf{s}_{d_1}, \mathbf{s}_{d_2}, \dots, \mathbf{s}_{d_k})$

- 1 **initialize** iteration $I = \{1, 2, \dots, N\}$, where N is the total number of raw melt pool images
- 2 **for each** $i \in I$ **do**
- 3 Convert raw image to grayscale image: $I_{gray_i} \leftarrow I_{raw_i}$
- 4 Find the maximum intensity of the image: $I_{MI_i} \leftarrow \max(I_{gray_i}(:))$
- 5 Perform segmentation to detect the plume: $\mathcal{S}_i \leftarrow (I_{MI_i} \times \mathcal{T}_{\min} \leq I_{gray_i} \leq I_{MI_i} \times \mathcal{T}_{\max})$
- 6 Define a structuring elements SE_{close} and SE_{open} for morphological operations
- 7 Implement morphological operations on the segmented image $\mathcal{S}_i$
- 8 Morphological closing using structural element SE_{close} : $I_{CE_i} \leftarrow \mathcal{F}(SE_{close}, r_c)$
- 9 Morphological opening operation to eliminate edges around melt pool: $I_{MP_i} \leftarrow \mathcal{F}(SE_{open}, r_o)$
- 10 Compute the centroid $\mathbf{g}_{\mathcal{S}_i}$ for the segmented melt pool image I_{MP_i}
- 11 Apply the k -means clustering algorithm to categorize the centroids $\{\mathbf{g}_{\mathcal{S}_i}\}_{i=1}^N$ into k clusters: $(\mathbf{s}_{d_1}, \mathbf{s}_{d_2}, \dots, \mathbf{s}_{d_k}) = k\text{-means}(\{\mathbf{g}_{\mathcal{S}_i}\}_{i=1}^N, k)$, where clusters $(\mathbf{s}_{d_1}, \mathbf{s}_{d_2}, \dots, \mathbf{s}_{d_k})$ represents a group of melt pool images moving in the same scan direction
- 12 **return** < Clusters of spatial coordinate points $(\mathbf{s}_{d_1}, \mathbf{s}_{d_2}, \dots, \mathbf{s}_{d_k})$ >

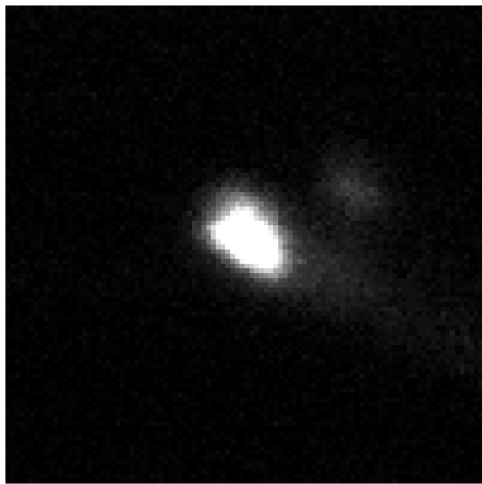
SPI – Algorithm and Performance



SPI performance on raw melt pool images
(succeeds in identifying the scan pattern
across different parts)

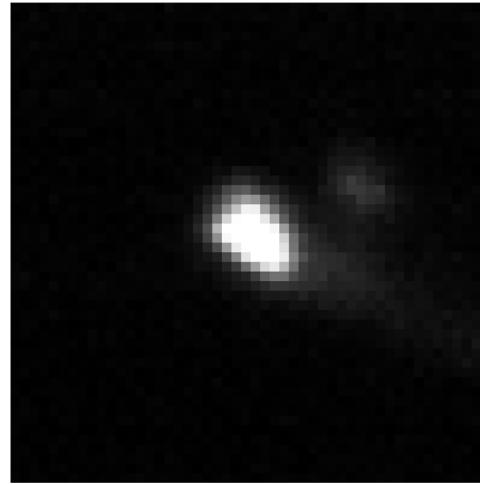
Mainstream Notion of Privacy – Pixelation

- Blurs image and obscure details

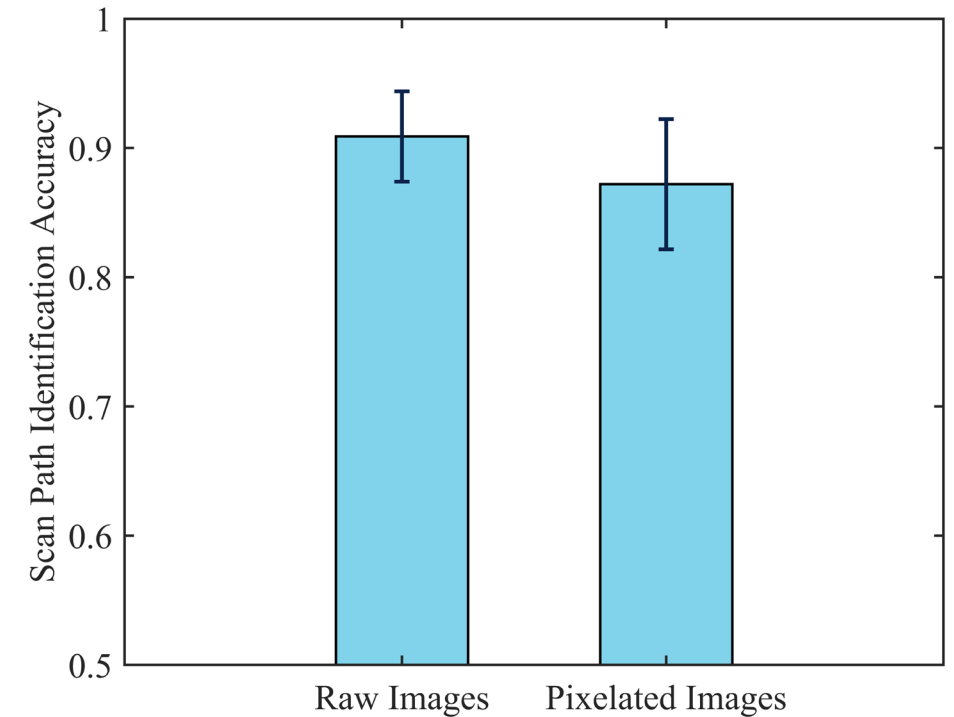


Original melt pool image

Pixelation
→
Noise
injection



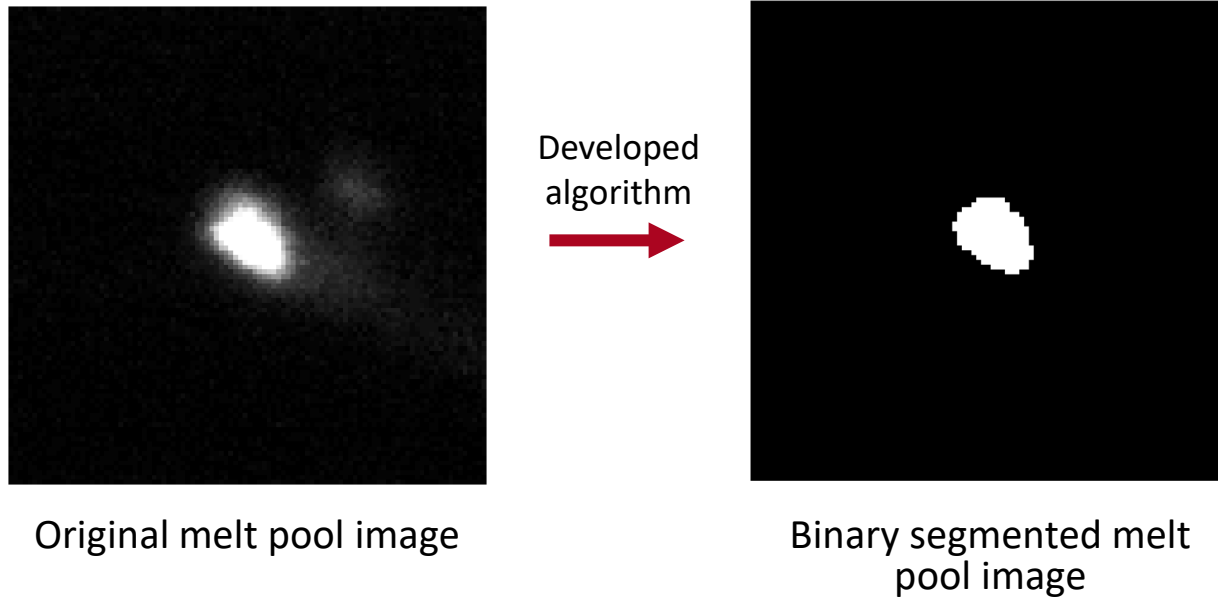
Pixelated melt pool image



SPI performance on raw and pixelated images
(succeeds in identifying the scan pattern across different parts)

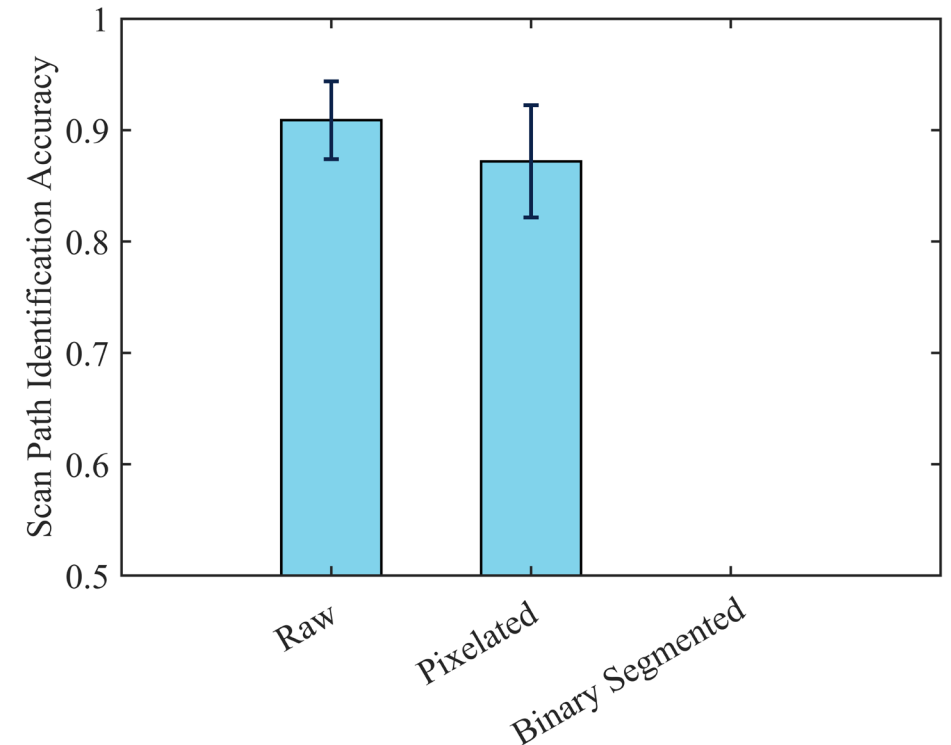
Privacy Mechanism 1 – Binary Segmentation

- Prevents leakage of scanning path information
- Preserves geometric descriptor of melt pool



Masking the plume information is beneficial for privacy preservation.

Is it sufficient?



SPI performance on binary segmented images
(fails to identify the scan pattern)

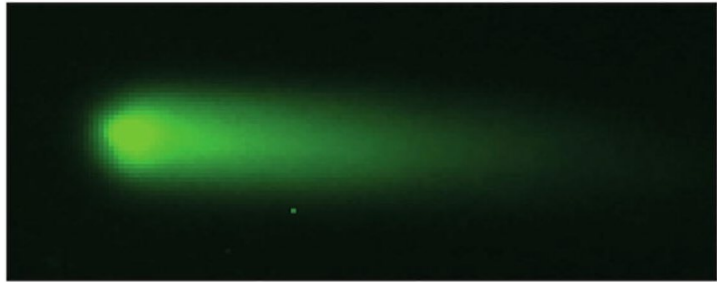
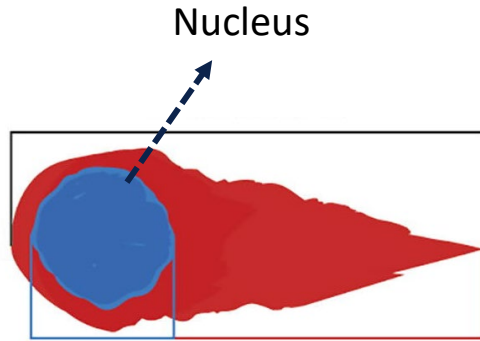
Scanning Path Reconstruction Attack (SPRA)

Input: Set of **binary segmented melt pool images** with randomized spatial coordinates $\{I_{BW\pi(i)}, \mathbf{c}_{\pi(i)}\}_{i=1}^N$, interval angle α , clusters of scan directions k

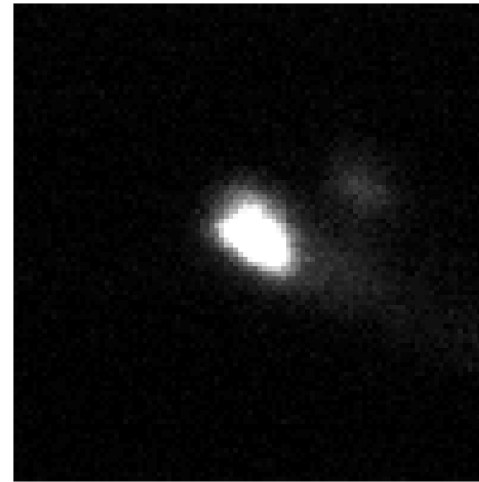
Output: Clusters of spatial coordinate points $(\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k)$

- 1 **initialize** iteration $I = \{1, 2, \dots, N\}$, where N is the total number of melt pool images
- 2 Initial sum of centroids $\boldsymbol{\delta} \leftarrow (0, 0)$
- 3 **for each** $i \in I$ **do**
- 4 Compute the centroid $\mathbf{g}_i$ for the segmented melt pool image $I_{BW\pi(i)}$, $\mathbf{g}_i = \left(\frac{1}{|\mathcal{R}_i|} \sum_{(x,y) \in \mathcal{R}_i} x, \frac{1}{|\mathcal{R}_i|} \sum_{(x,y) \in \mathcal{R}_i} y \right)$, where $\mathcal{R}_i = \{(x, y)_1, (x, y)_2, \dots\}$ is the set of the coordinates of the foreground pixels in $I_{BW\pi(i)}$
- 5 Update the sum of centroids, $\boldsymbol{\delta} \leftarrow \boldsymbol{\delta} + \mathbf{g}_i$
- 6 Compute the mean of all centroids to identify the reference point $\mathbf{r} = \left(\frac{1}{N} \right) \boldsymbol{\delta}$
- 7 **for each** $i \in I$ **do**
- 8 **initialize** Reference direction = +ve X-axis
- 9 Detect the set of coordinates of the border pixels: $P_i = \{\mathbf{p}_1, \mathbf{p}_2, \dots\}$ for $I_{BW\pi(i)}$
- 10 **for each** angle $\theta_{MF_j} \in \{0^\circ, \alpha, 2\alpha, \dots, 360^\circ - \alpha\}$ from Reference direction **do**
- 11 Compute the distance of the border pixel $\mathbf{p}_j \in P_i$ from $\mathbf{r}$ along direction θ_{MF_j} : $d_{i, \theta_{MF_j}} = \|\mathbf{p}_j - \mathbf{r}\|$
- 12 Create the motion feature vector $\mathbf{m}_i = (d_{i, 0^\circ}, d_{i, \alpha}, \dots, d_{i, 360^\circ - \alpha})$
- 13 Apply the k -means clustering algorithm to categorize the motion feature vectors $\{\mathbf{m}_i\}_{i=1}^N$ into k clusters: $(\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k) = k\text{-means}(\{\mathbf{m}_i\}_{i=1}^N, k)$, where clusters $(\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k)$ represents a group of melt pool images moving in the same scan direction
- 14 **return** < Clusters of spatial coordinate points $(\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k)$ >

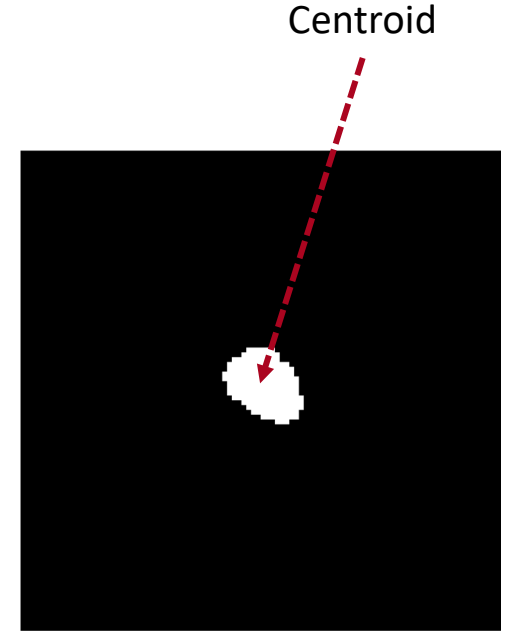
Background – Utilizing Process Knowledge



A comet



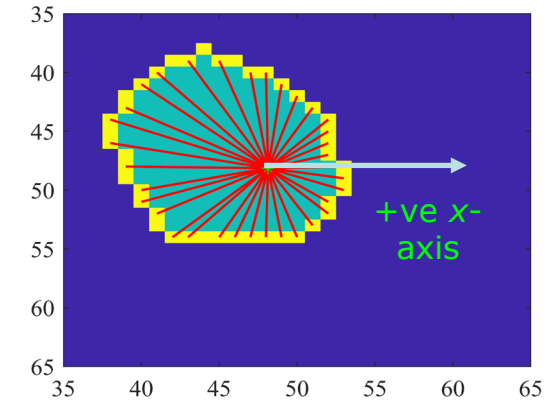
Original melt pool image



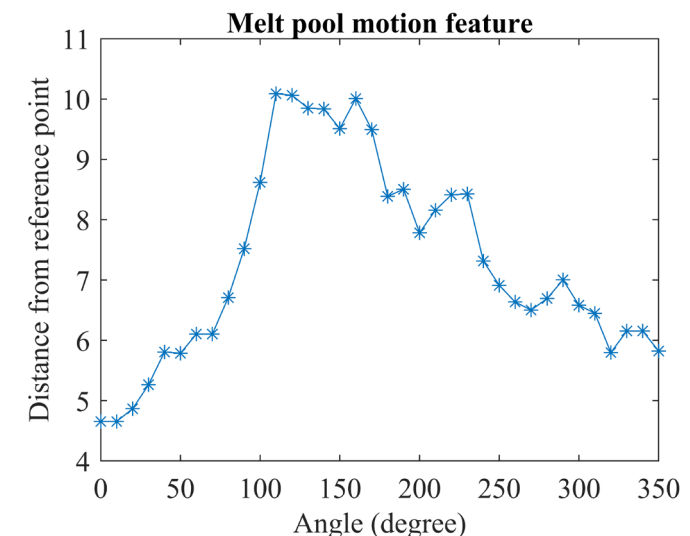
Binary segmented melt pool image

Scanning Path Reconstruction Attack (SPRA)

- Identify the coordinates of the centroids of the melt pool regions & compute their mean.
 - This mean value is used as the **reference point**.
- For each melt pool image – measure the distance of the border pixels from the **reference point** at an interval of 10° starting from the direction of the +ve x-axis and form a vector of **motion features**.
- Use **Machine Learning (ML) algorithm** to group the motion feature vectors into clusters.



+ve x-axis as reference direction & mean of the centroids as reference point (proposed SPRA)



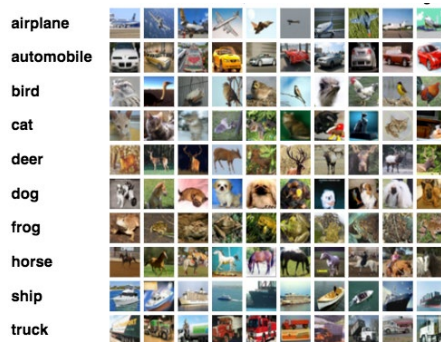
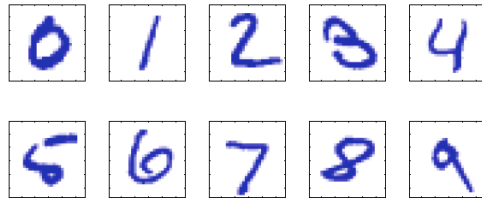
Motion features under proposed SPRA

Machine Learning Overview

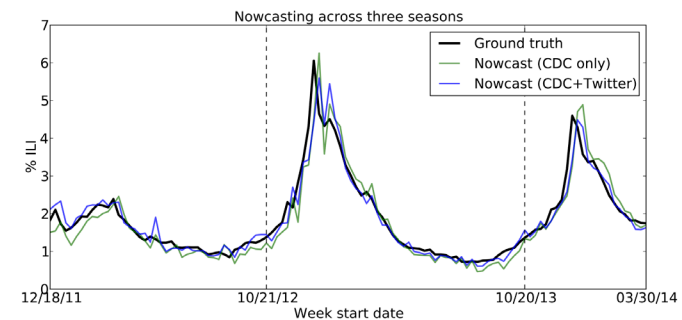
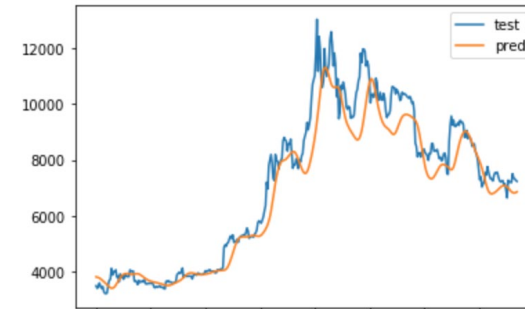
■ Supervised learning

- Has a baseline understanding of the correct output
- Labeled training data provided (X = data, Y = labels (or target))
- Examples: conventional regression/classification problems fall into this category

Classification
(Y takes discrete set of values)



Regression
(Y is continuous valued)



Machine Learning Overview

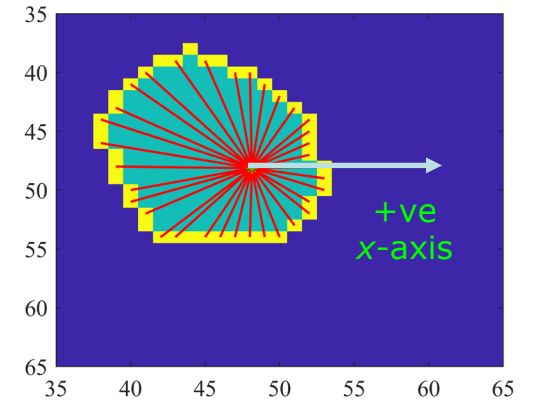
- Unsupervised learning
 - No labels provided;
 - The goal is to learn underlying features/patterns in the data
 - **Examples:** clustering, density estimation, low-dimensional representation of data
- Reinforcement learning
 - Learning by interacting with the environment
 - The goal is to design optimal actions that maximize a reward
 - **Examples:** robotics, control systems, recommendation systems, resource management

Let's go back to scan the path identification problem

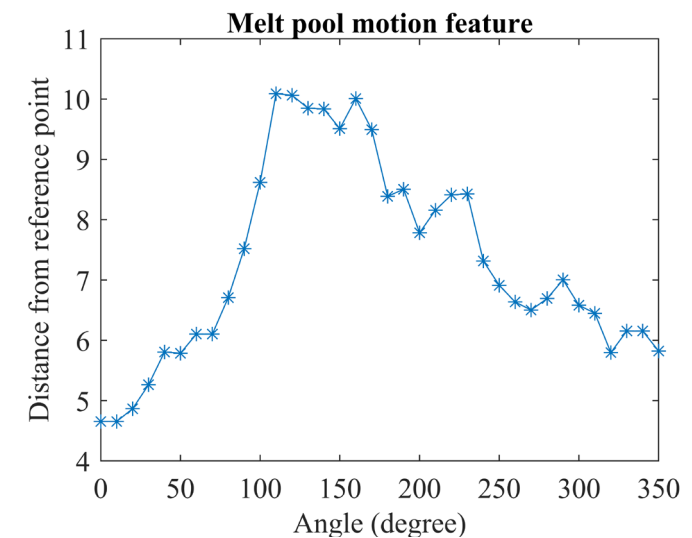


Scanning Path Reconstruction Attack (SPRA)

- Identify the coordinates of the centroids of the melt pool regions & compute their mean.
 - This mean value is used as the **reference point**.
- For each melt pool image – measure the distance of the border pixels from the **reference point** at an interval of 10° starting from the direction of the +ve x-axis and form a vector of **motion features**.
- Use **Machine Learning (ML) algorithm** to group the motion feature vectors into clusters.
 - Do we have **labeled** data?
 - What is the **goal**?



+ve x-axis as reference direction & mean of the centroids as reference point (proposed SPRA)

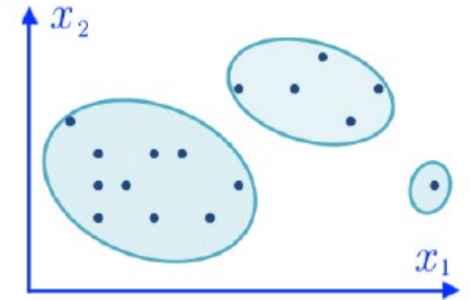


Motion features under proposed SPRA

Unsupervised Machine Learning - Clustering

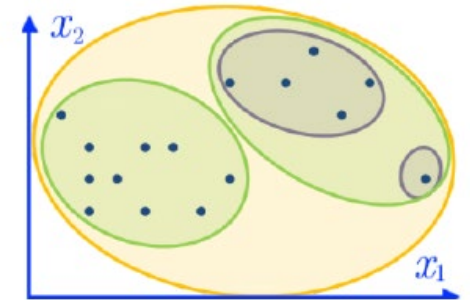
■ Partitioning approach

- Construct various partitions in X space with some optimization criteria
- Typical Methods: k-means, k-medoids



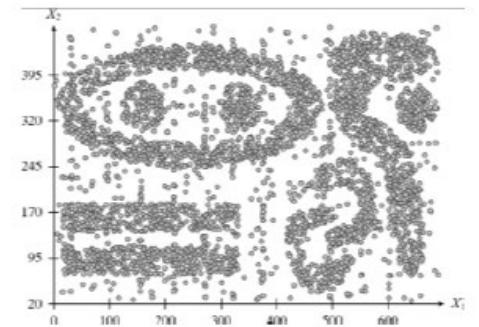
■ Hierarchical approach

- Decompose X space hierarchically with some optimization criteria
- Typical Methods: divisive (top-down), agglomerative (bottom-up)

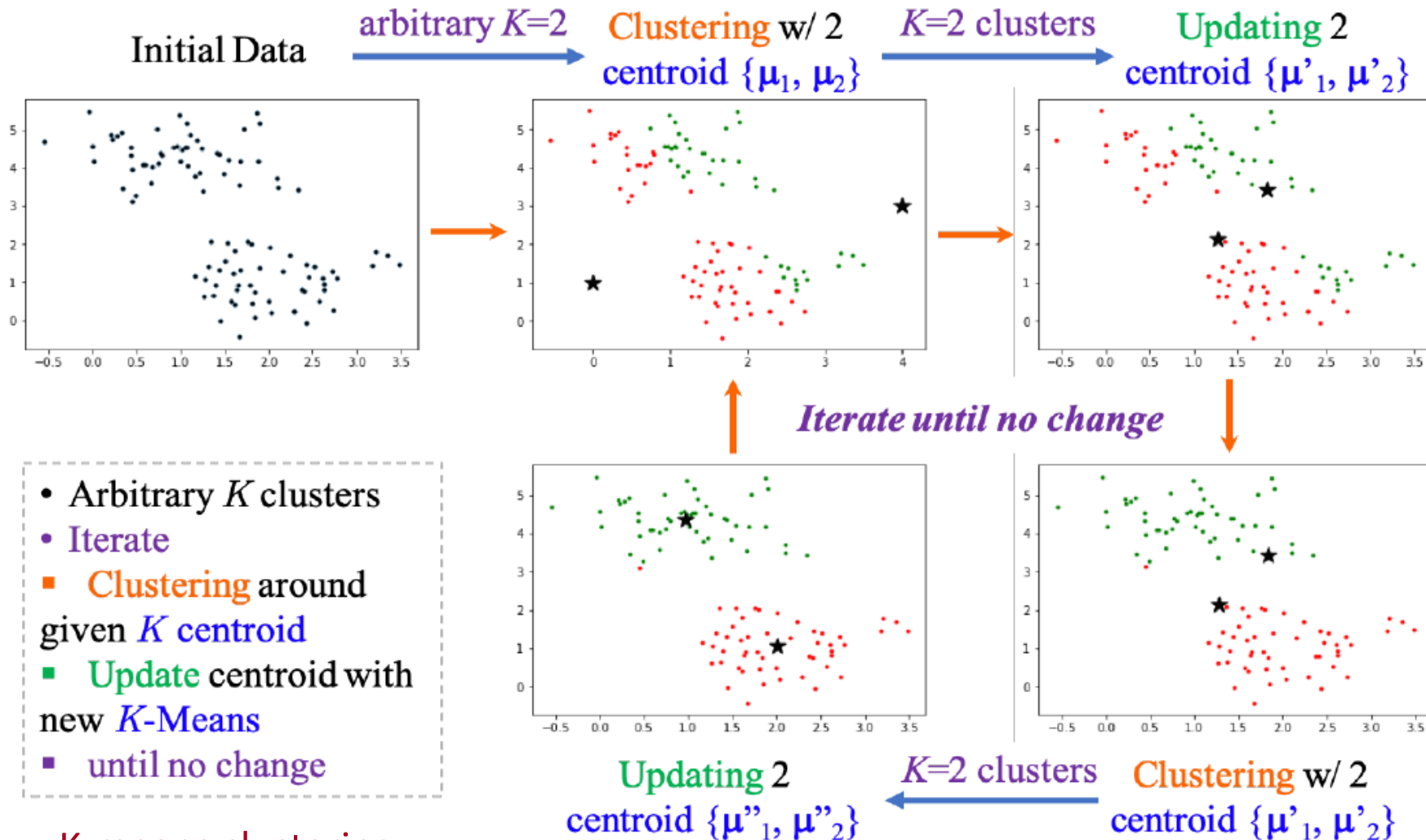


■ Density-based approach

- Estimate density function
- Cluster X space according to density similarity



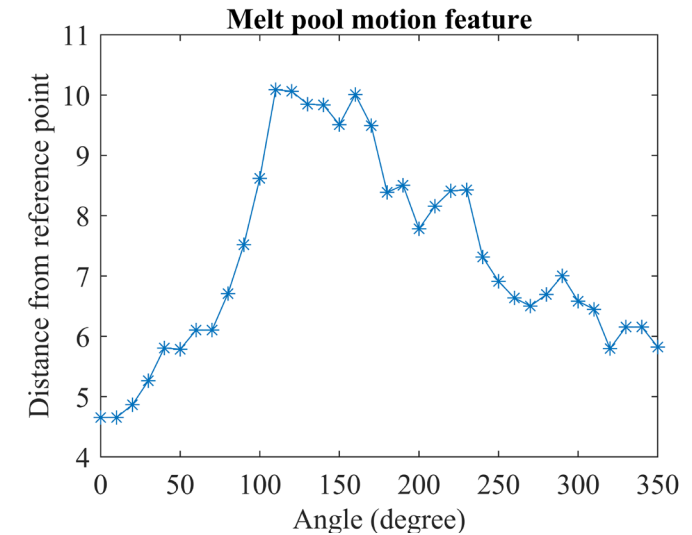
K-means clustering – Graphical Illustration



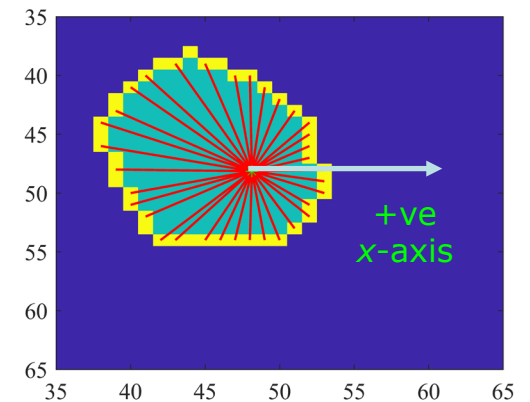
K-means clustering
algorithm

Scanning Path Reconstruction Attack (SPRA)

- Identify the coordinate of the centroids of the melt pool regions & compute their mean.
 - This mean value is used as **reference point**.
- For each melt pool image – measure the distance of the border pixels from the **reference point** at an interval of 10° starting from the direction of the +ve x-axis and form a vector of **motion features**.
- Use **k-means clustering algorithm** to group the motion feature vectors into two clusters.
 - The melt pools that are moving in the same direction are expected to have similar motion features and are expected to be clustered together.



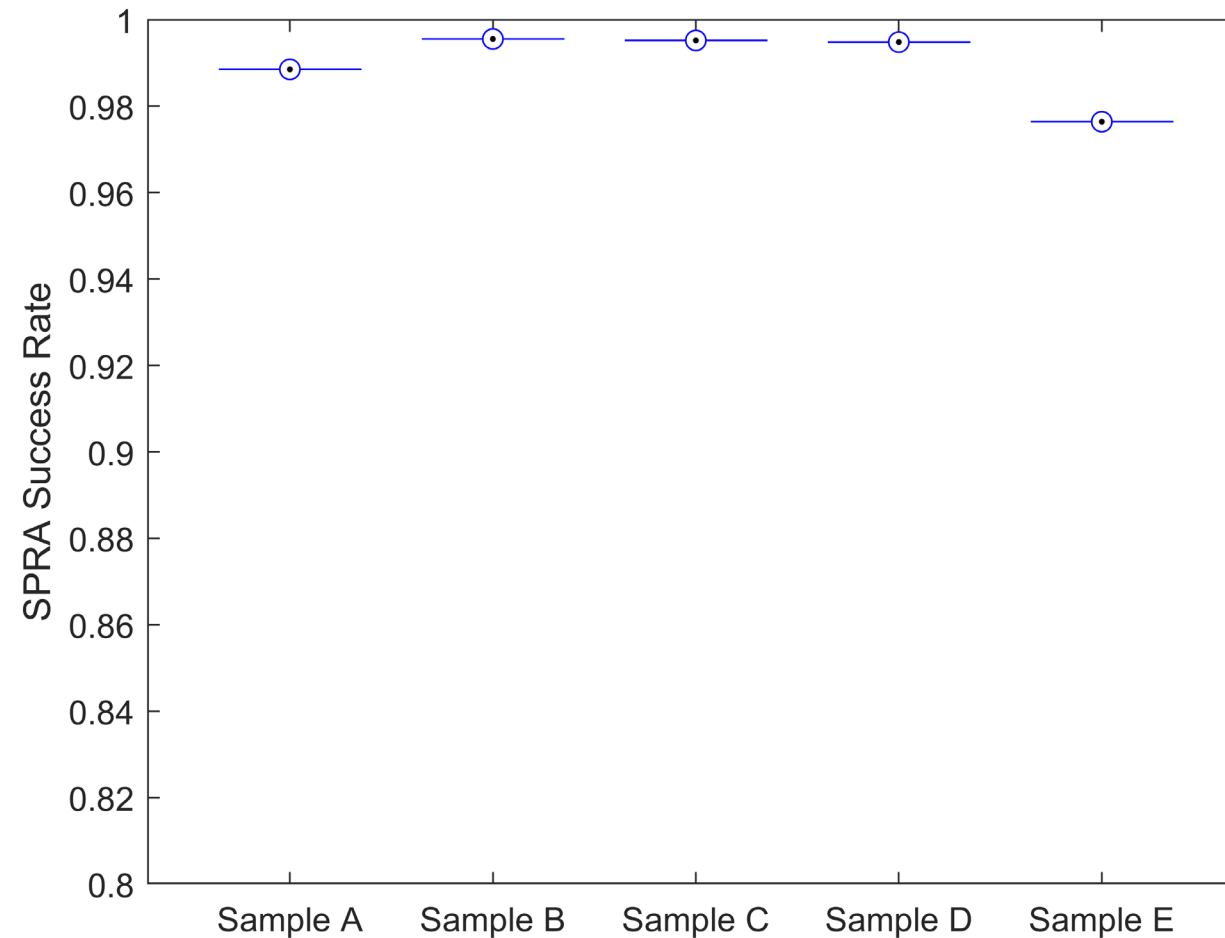
Motion features under proposed SPRA



+ve x-axis as reference direction & mean of the centroids as reference point (proposed SPRA)

SPRA Algorithm – Performance Against PM-1

Evaluation of SPRA success rate against PM-1



Privacy Mechanism 2 – Image Perturbation

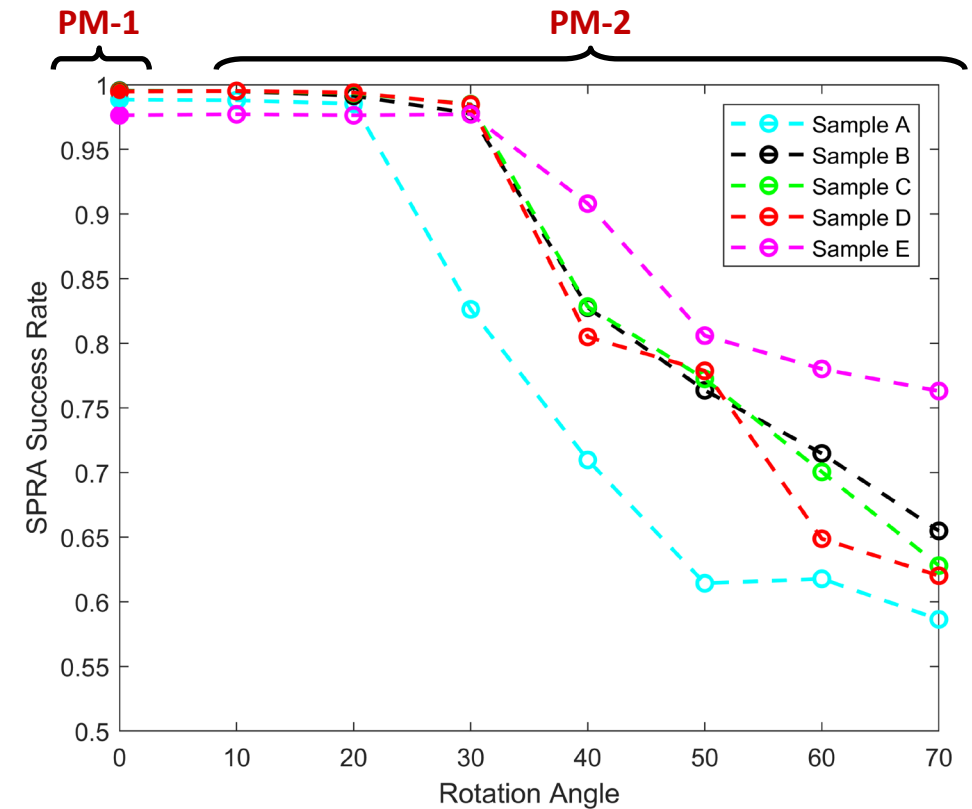
Input: Shuffled **binary segmented melt pool images** with coordinates $\{I_{BW_{\pi(i)}}, \mathbf{c}_{\pi(i)}\}_{i=1}^N$,
rotation angle range θ_R

Output: Perturbed binary segmented images with coordinates $\{I_{BW_{\pi(i)}}^R, \mathbf{c}_{\pi(i)}\}_{i=1}^N$

- 1 **initialize** iteration $I = \{1, 2, \dots, N\}$, where N is the total number of binary segmented melt pool images
 - 2 **for each** $i \in I$ do
 - 3 Sample a random rotation angle θ_{R_i} from a Uniform distribution within the range $[-\theta_R, \theta_R]$: $\theta_{R_i} \sim U(-\theta_R, \theta_R)$
 - 4 Apply rotation θ_{R_i} to the binary segmented image $I_{BW_{\pi(i)}}$: $I_{BW_{\pi(i)}}^R \leftarrow \text{Rotate}(I_{BW_{\pi(i)}}, \theta_{R_i})$
 - 5 **return** < Perturbed and shuffled binary segmented images with coordinates $\{I_{BW_{\pi(i)}}^R, \mathbf{c}_{\pi(i)}\}_{i=1}^N$ >
- 

Privacy Mechanism 2 – Effectiveness

- Under Privacy Mechanism-2, the binary segmented melt pool images are **perturbed with $[-\theta^\circ \sim \theta^\circ]$ rotation** before sharing.
- The SPRA success rate was investigated under different degree of rotations - $\pm 10^\circ$ to $\pm 70^\circ$ with an increment of 10° ; across **five samples** and **30 observations per sample**.
 - SPRA success rate is not much affected up to $\pm 20^\circ$ rotation.
 - SPRA success rate declines after $\pm 30^\circ$ rotation.
 - The decreasing trend is consistent across all the samples.



Effectiveness of privacy mechanism 2 against SPRA across five samples

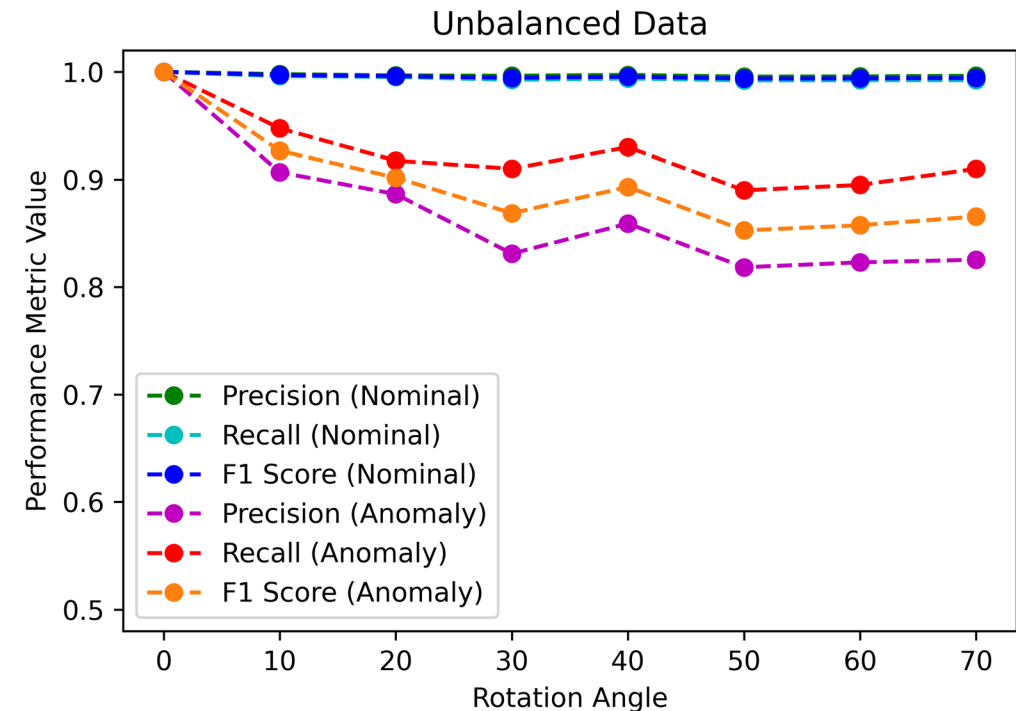
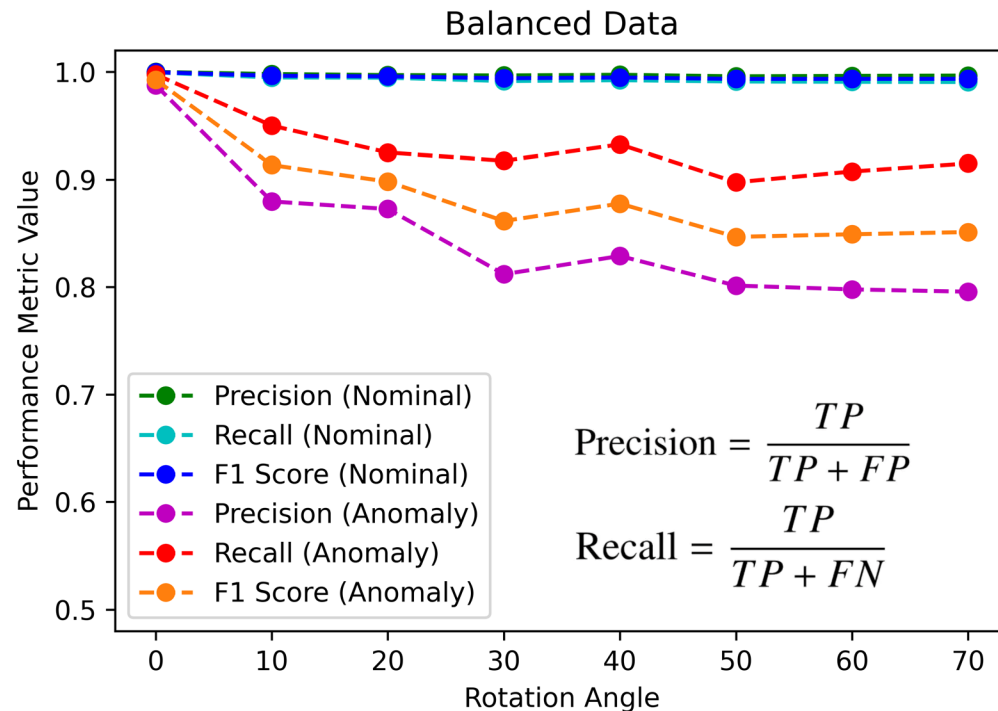
Evaluation of Data Utility

- Data Utility: Performance of ML classifier in predicting anomaly
 - Two labels: Nominal (non-defective) and Anomaly
 - 10330 meltpool images
 - **Random Forest** classifier
- Unbalanced dataset (9930 nominal data, 400 anomaly data)
 - Data balancing using **SMOTE** (Synthetic Minority Over-sampling Technique)



Evaluation of Data Utility

- Data utility can be preserved with both privacy mechanisms
- Slightly higher recall for anomaly in balanced data; slightly higher precision in unbalanced data



Significance of Results

- Presents the proof of concept for **information leakage** from raw melt pool image data sharing and proactively developed a **novel adversarial attack** to discover confidential product design information
- Introduces two physics-informed and security-aware **data-sharing strategies** for metal AM process data and provides a foundation to investigate **process information leakage**





Physics-informed and Data-driven Methods for Developing **Detective** and Preventive Cyber-physical Defenses

Security-aware Process Monitoring

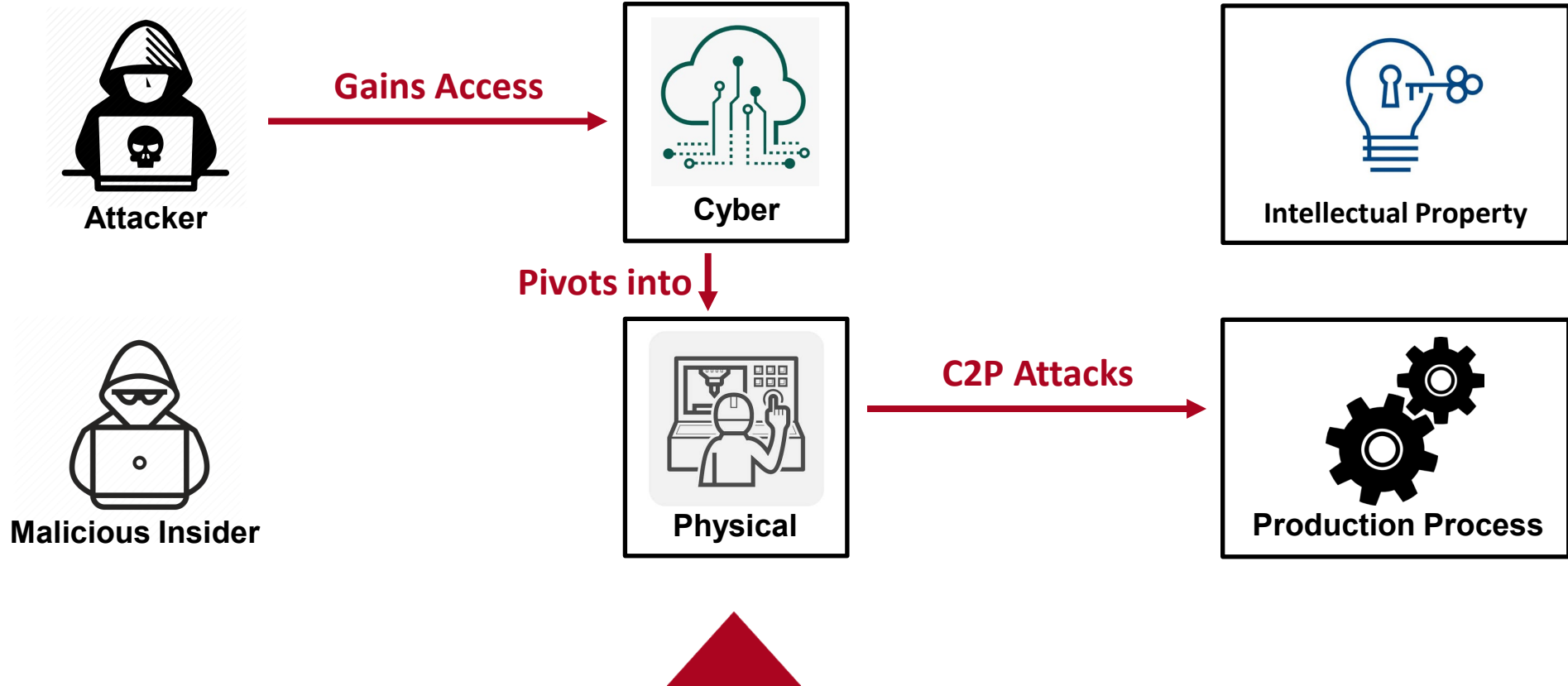
Habibor Rahman

Assistant Professor, Mechanical Engineering
University of Massachusetts Dartmouth

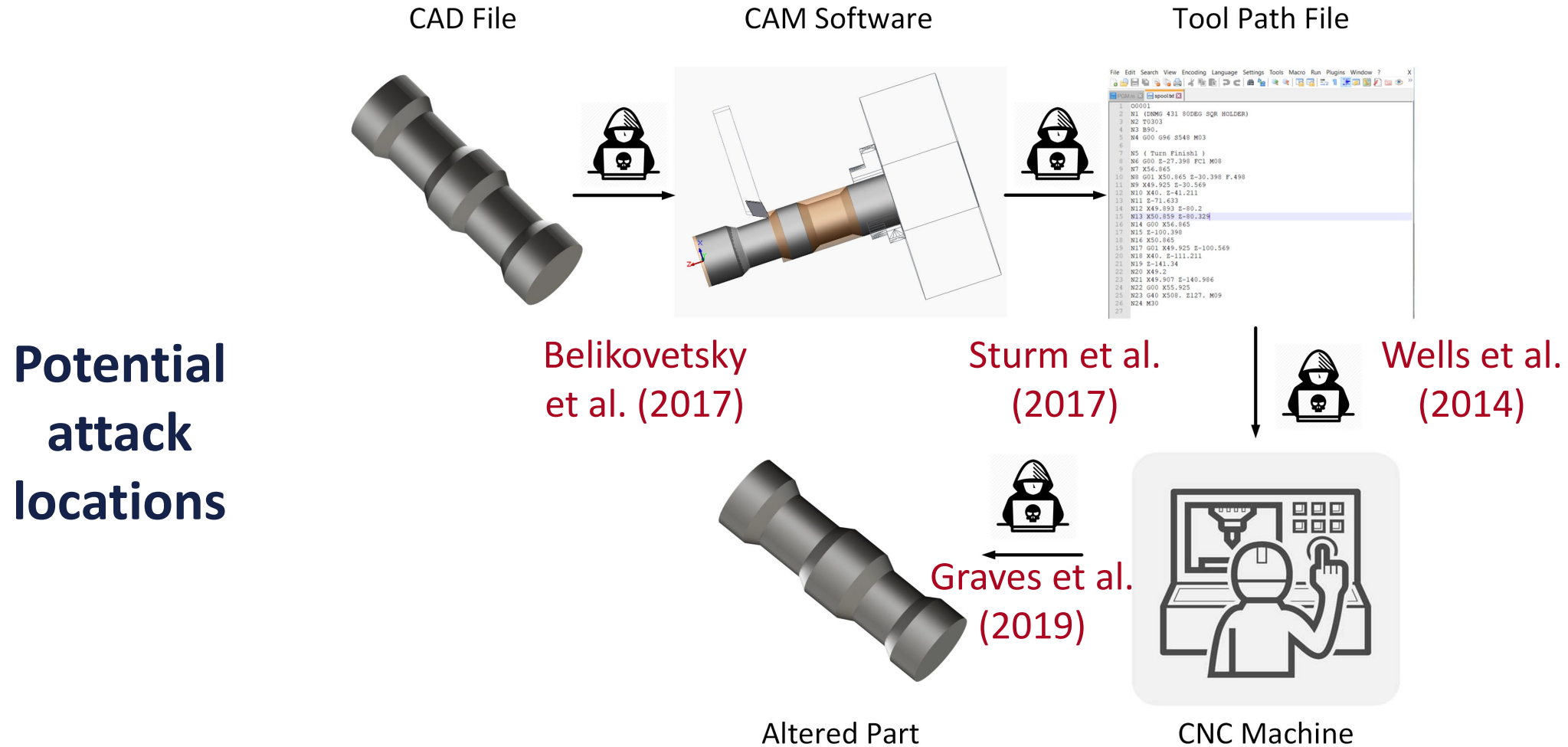


Motivation - Product Oriented C2P Attacks

Attacks on manufacturing can be described by the domain that the attack is targeting/influencing (cyber or physical domain) and the domain that is being affected/victimized (cyber or physical domain)

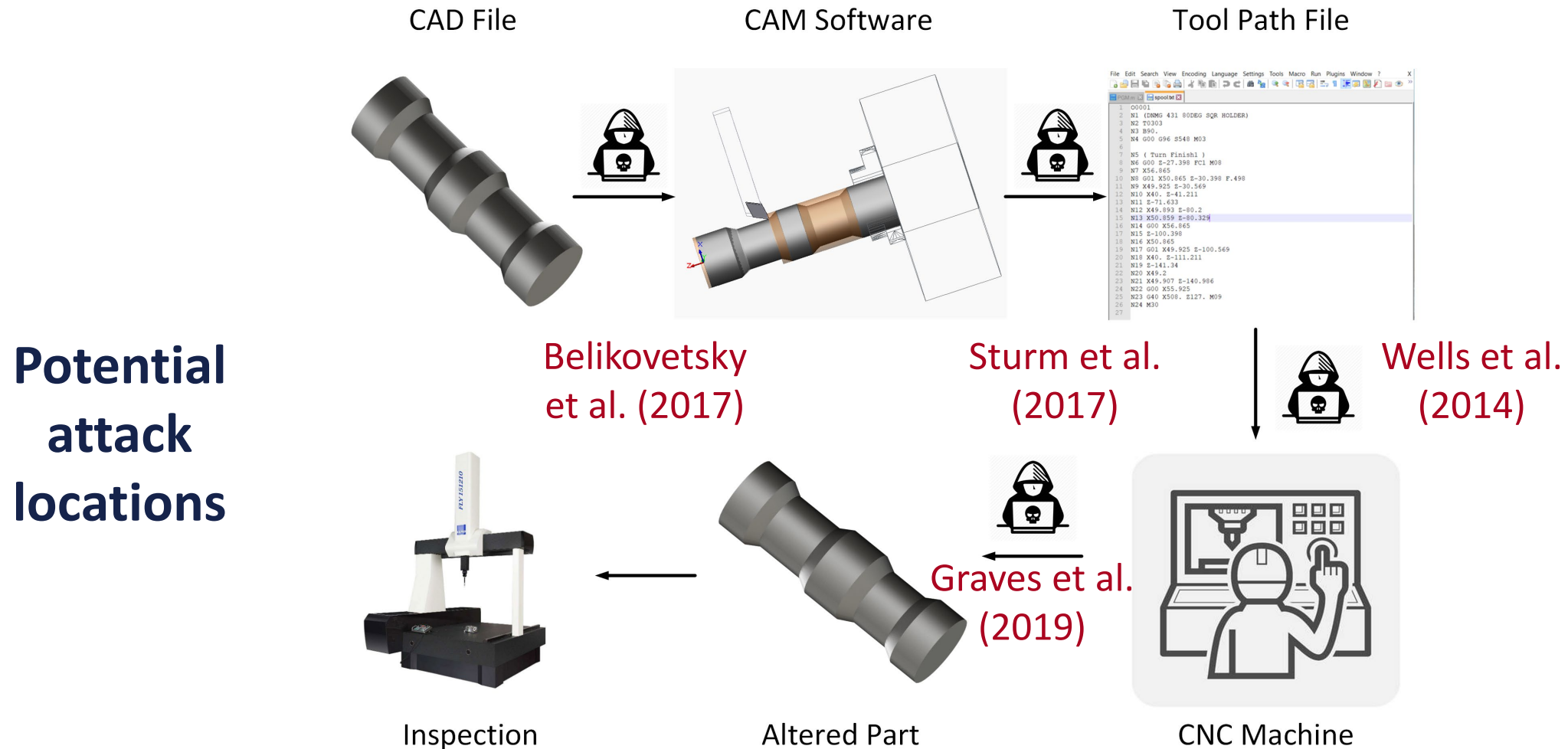


Product Oriented C2P Attacks - Locations



1. Belikovetsky S, Yampolskiy M, Toh J, Gatlin J, Elovici Y. Dr0wned – Cyber-physical attack with additive manufacturing. 11th USENIX Work. 2017.
2. Sturm, L. D., Williams, C. B., Camelio, J. A., White, J., & Parker, R. (2017). Cyber-physical vulnerabilities in additive manufacturing systems: A case study attack on the. STL file with human subjects. Journal of Manufacturing Systems, 44, 154-164.
3. Wells LJ, Camelio JA, Williams CB, White J. Cyber-physical security challenges in manufacturing systems. Manuf Lett 2014;2:74–7.
4. Graves LMG, Lubell J, King W, Yampolskiy M. Characteristic aspects of additive manufacturing security from security awareness perspectives. IEEE Access 2019;7: 103833–53.

Product Oriented C2P Attacks - Locations



1. Belikovetsky S, Yampolskiy M, Toh J, Gatlin J, Elovici Y. Dr0wned – Cyber-physical attack with additive manufacturing. 11th USENIX Work. 2017.
2. Sturm, L. D., Williams, C. B., Camelio, J. A., White, J., & Parker, R. (2017). Cyber-physical vulnerabilities in additive manufacturing systems: A case study attack on the. STL file with human subjects. Journal of Manufacturing Systems, 44, 154-164.
3. Wells LJ, Camelio JA, Williams CB, White J. Cyber-physical security challenges in manufacturing systems. Manuf Lett 2014;2:74–7.
4. Graves LMG, Lubell J, King W, Yampolskiy M. Characteristic aspects of additive manufacturing security from security awareness perspectives. IEEE Access 2019;7: 103833–53.

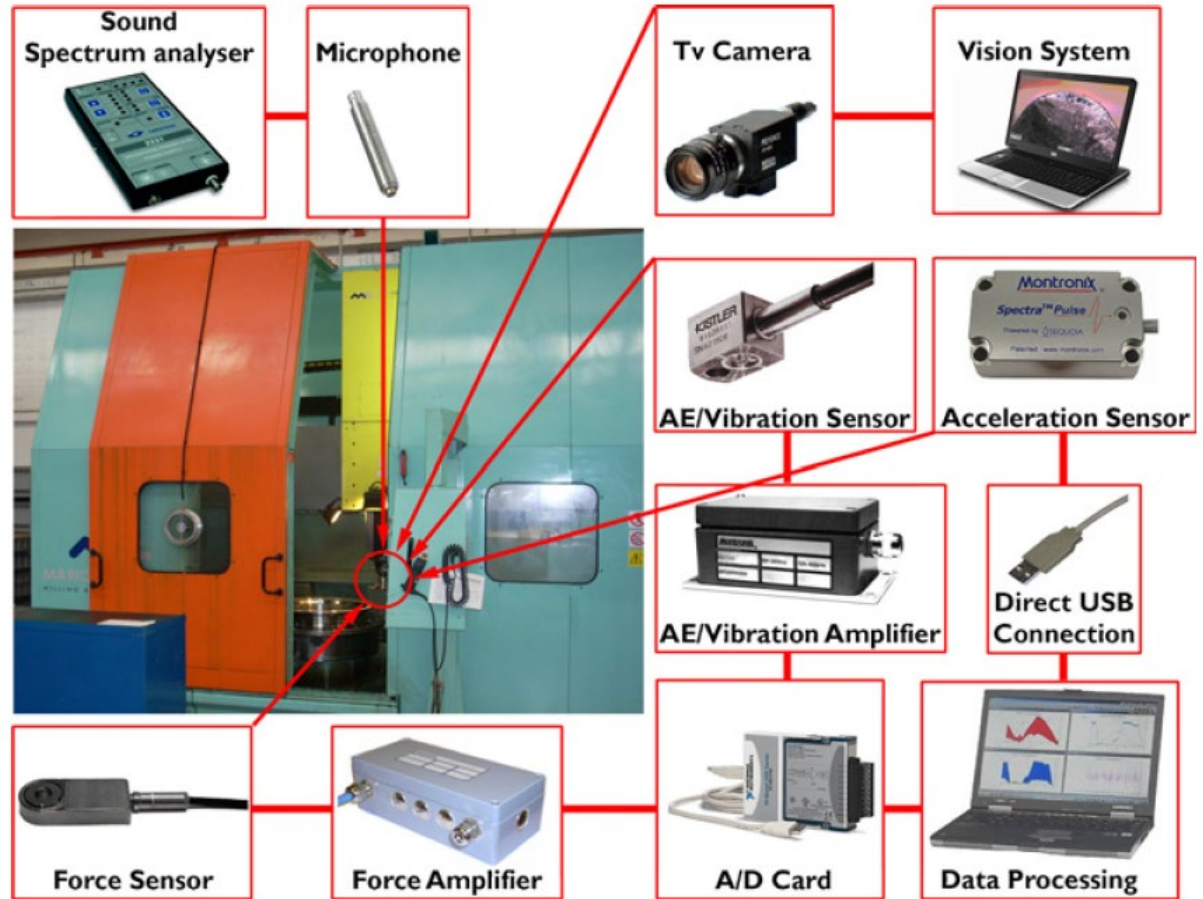
Inspection Systems in Machining - Offline

- Coordinate Measuring Machine
 - Measure precise dimensions, geometric tolerances, and surface quality
 - Optical Inspection System
 - Employ laser scanners or cameras to detect surface defects and dimensional inconsistencies
 - Ultrasonic Testing
 - Uses high-frequency sound waves for internal defect detection
 - Surface Roughness Testers
 - Measures surface finish to ensure compliance with design specifications
- 

Inspection Systems in Machining - Online

Common sensors and sensor systems

- Motor power and current
- Force and torque
 - Piezoelectric sensors (dynamic)
 - Strain gauges (static)
- Acoustic emission
- Vibration
- Temperature



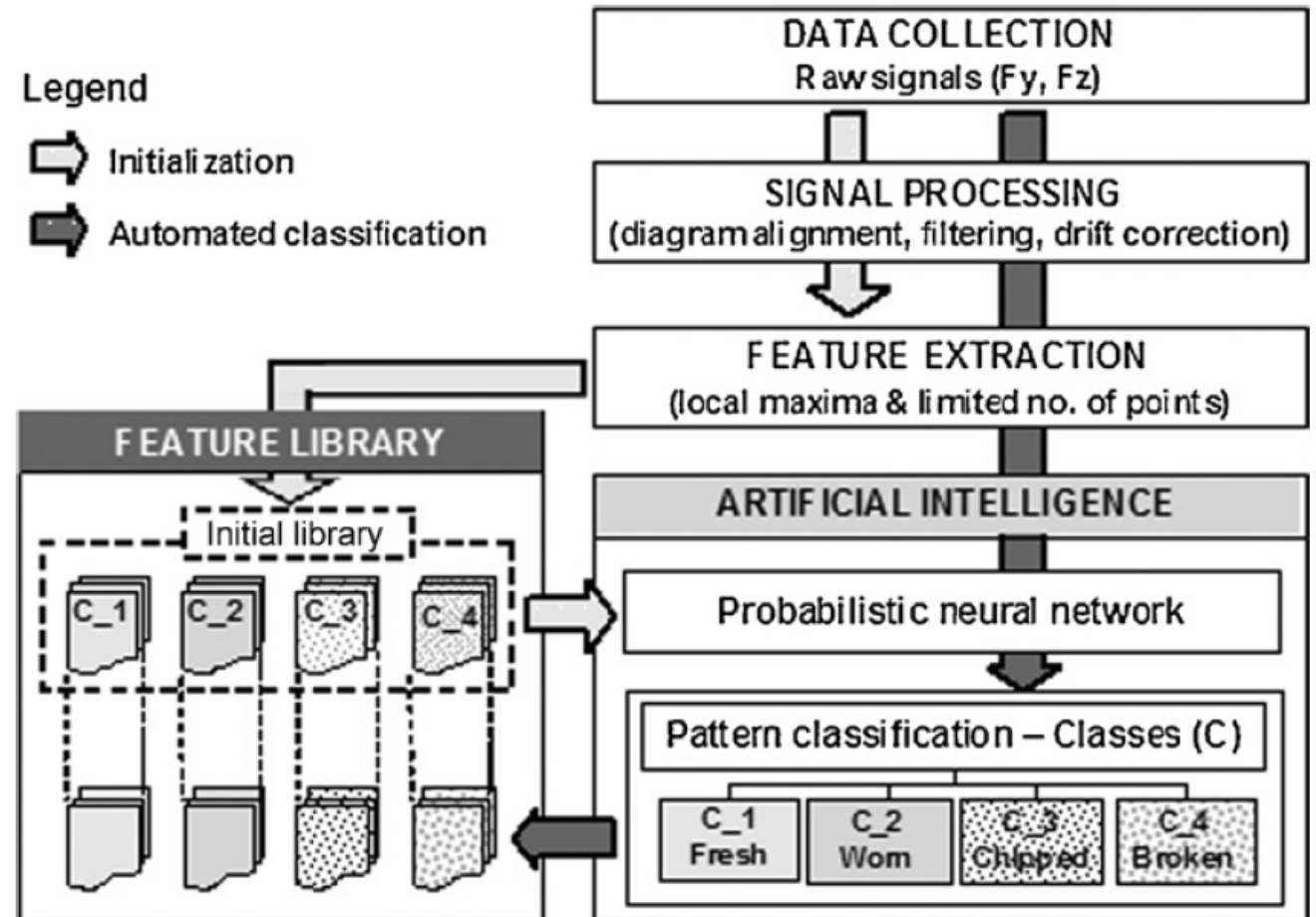
Inspection Systems in Machining - **Online**

Signal
preprocessing

Feature
extraction

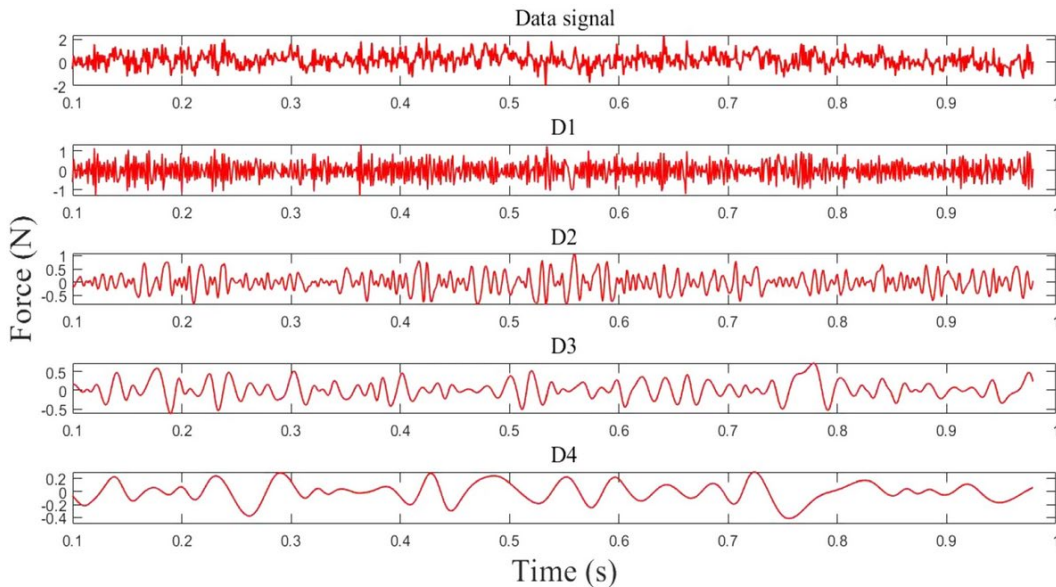
Signal feature
selection

Schematic of the tool condition recognition system



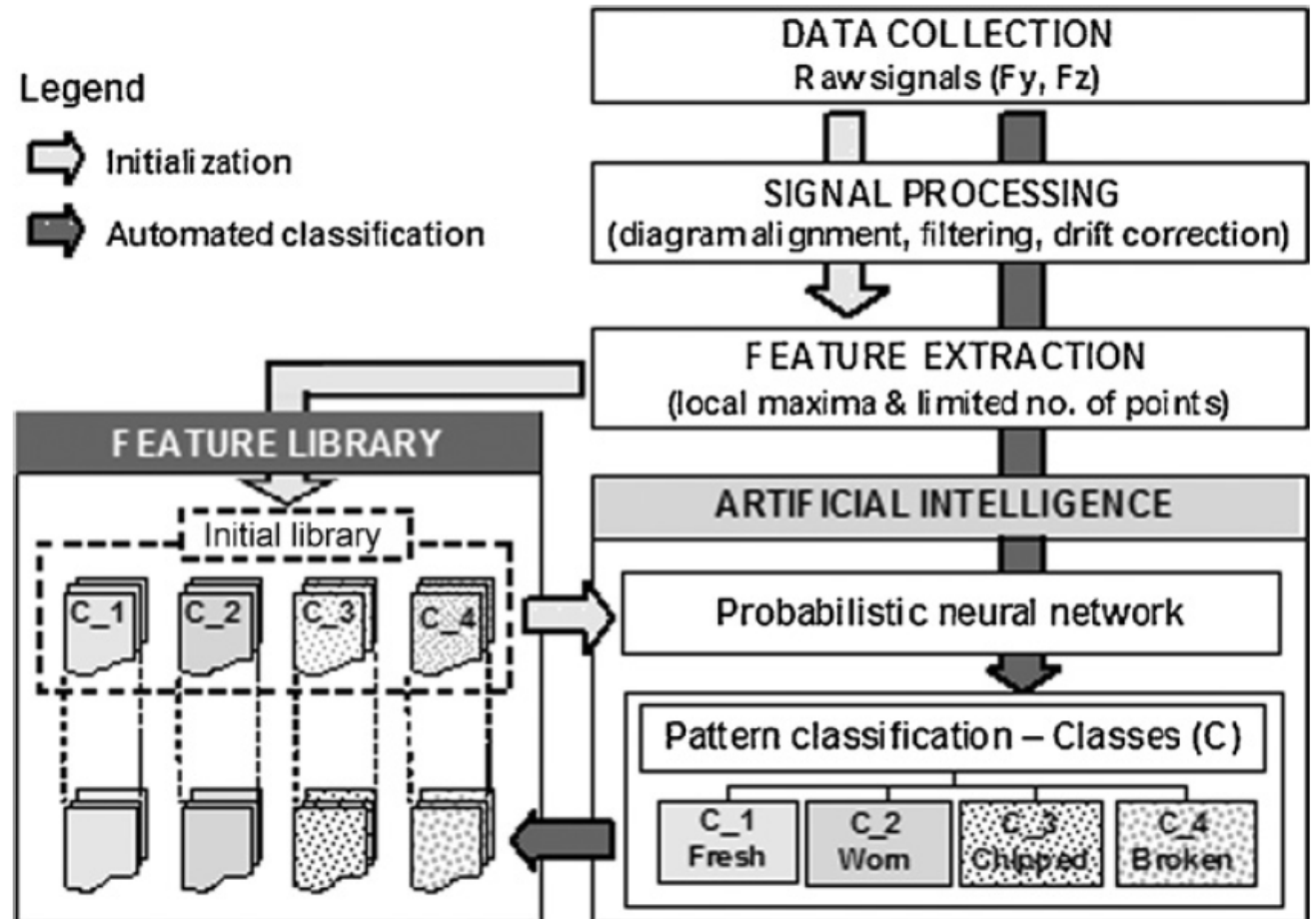
Inspection Systems in Machining - Online

Signal transformation and decomposition



Discrete Wavelet Transform (DWT) decomposes a signal and helps capture the overall trend (slow variation) and sharp changes in the signal

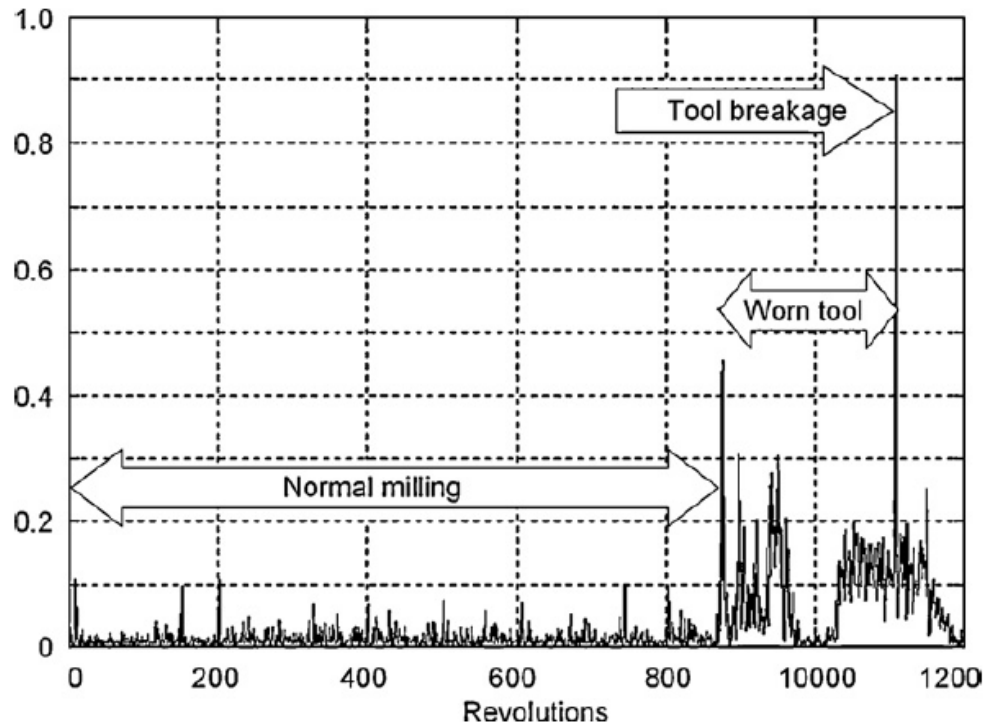
Schematic of the tool condition recognition system



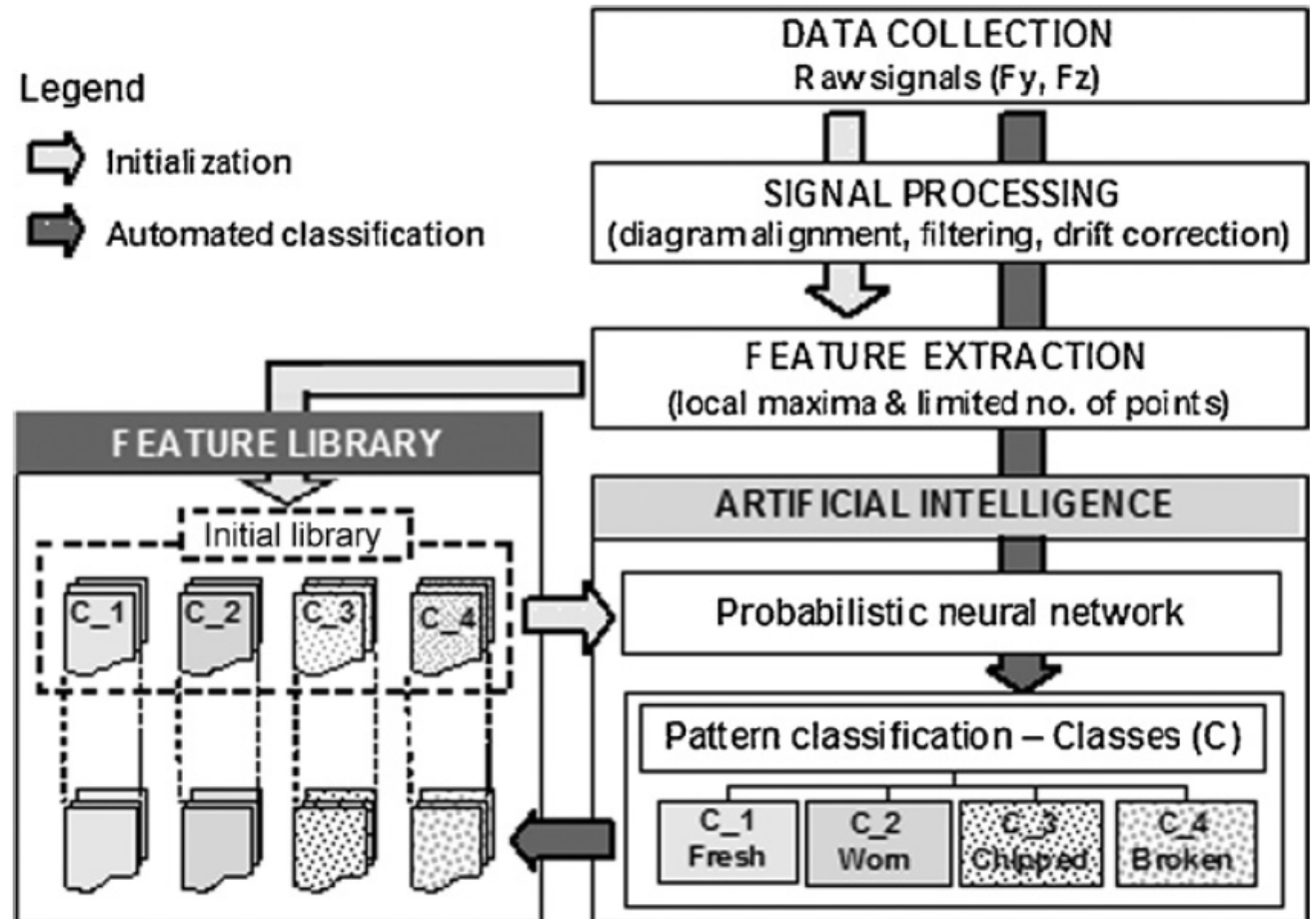
Source: doi:10.1016/j.cirp.2010.05.010

Inspection Systems in Machining - Online

Cutting force DWT during milling operation



Schematic of the tool condition recognition system



Traditional Inspection – Not Sufficient

Focus on traditional causes of variation such as **tool wear** and **chatter**

Post-production and in situ inspections can be **violated** in carefully designed attacks

Potential inadvertent mistakes leading to changes in product **geometry**

Evading post-production inspection

Knowledge of the **inspection plan**, the product's key quality characteristics (**KQCs**), and the **inspection procedure**

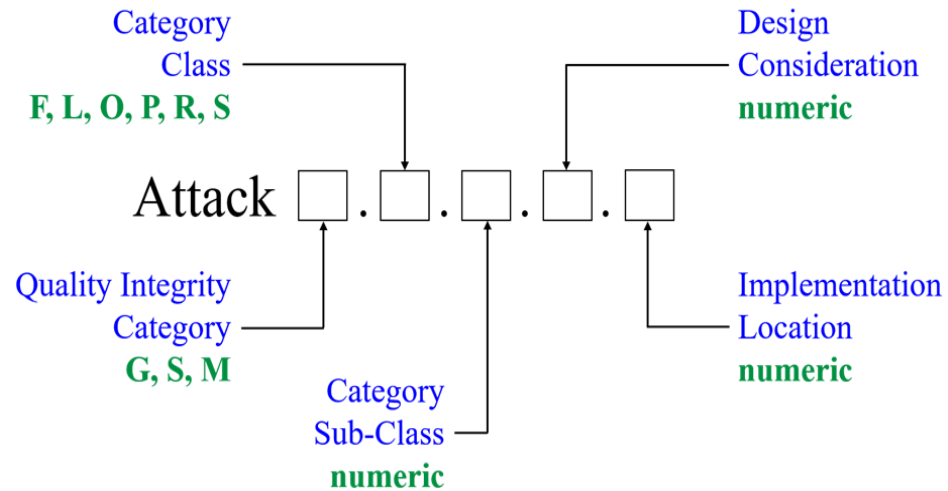
Evading in situ process monitoring

Knowledge of what physical **signal descriptors** will be monitored

Shafae, M. S., Wells, L. J., & Purdy, G. T. (2019). Defending against product-oriented cyber-physical attacks on machining systems. *The International Journal of Advanced Manufacturing Technology*, 105(9), 3829-3850.

Product Oriented C2P Attacks – Design and Designation

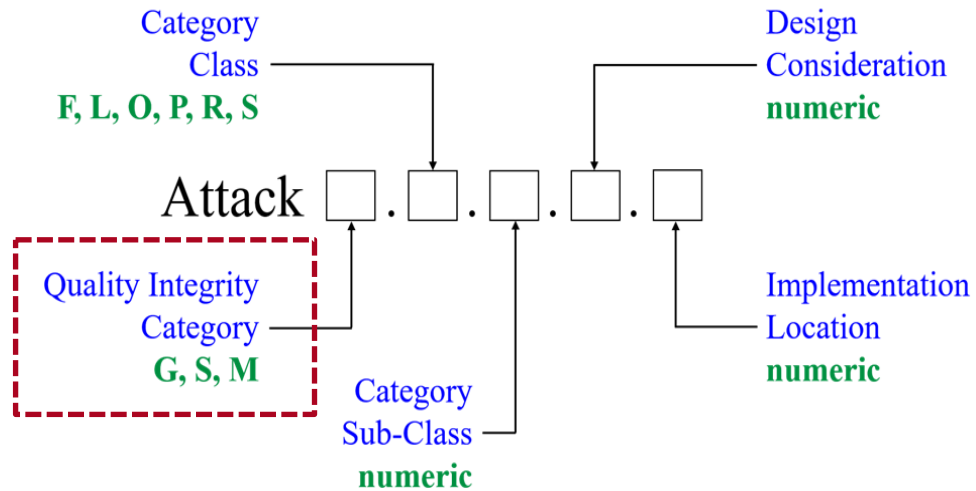
Attack Design and Designation Scheme



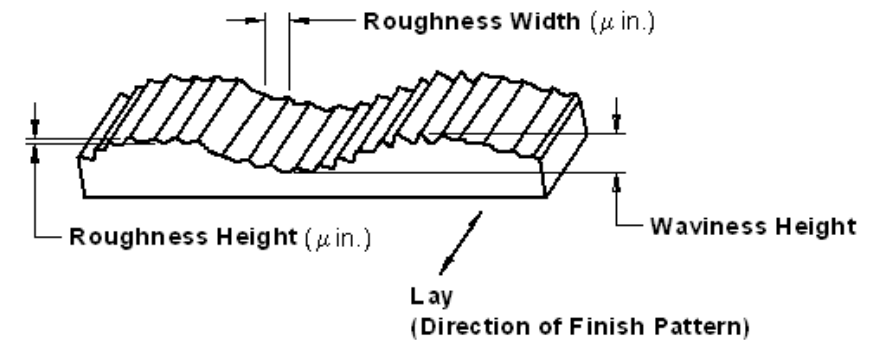
Reference. Shafae, M. S., Wells, L. J., & Purdy, G. T. (2019). Defending against product-oriented cyber-physical attacks on machining systems. *The International Journal of Advanced Manufacturing Technology*, 105, 3829-3850. <https://doi.org/10.1007/s00170-019-03805-z>

Product Oriented C2P Attacks – Design and Designation

Attack Design and Designation Scheme

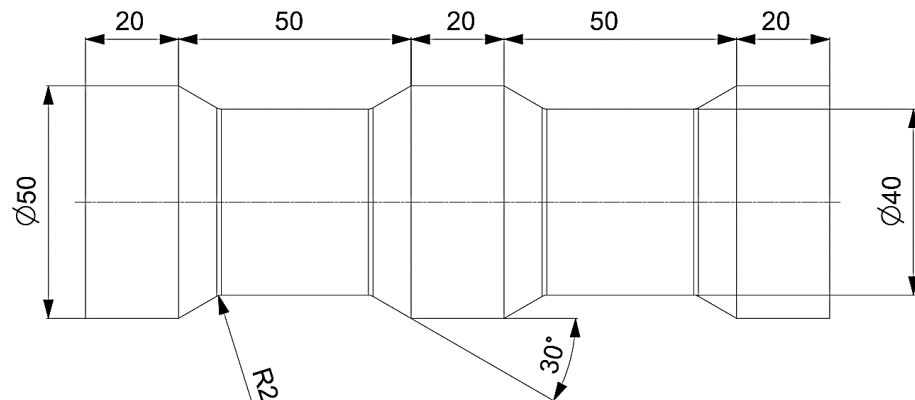


Surface Integrity

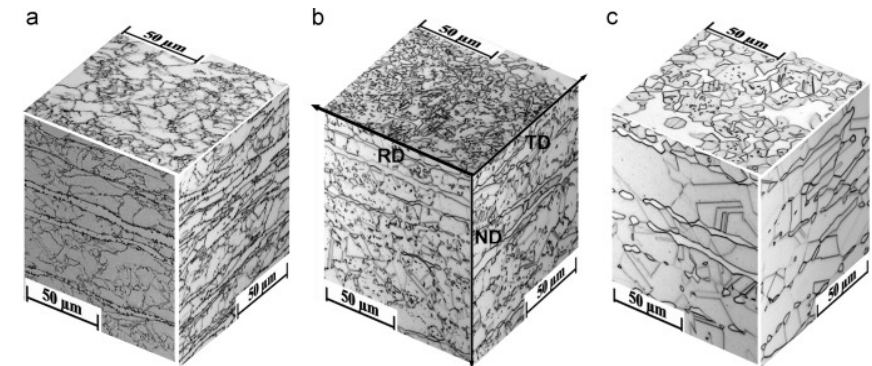


https://www.engineersedge.com/surface_finish.htm

Geometrical Integrity



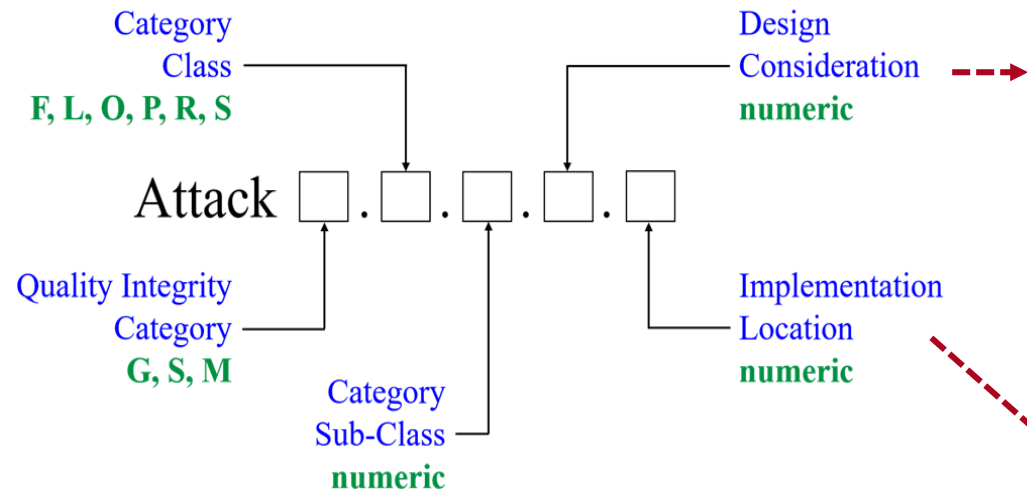
Material Integrity



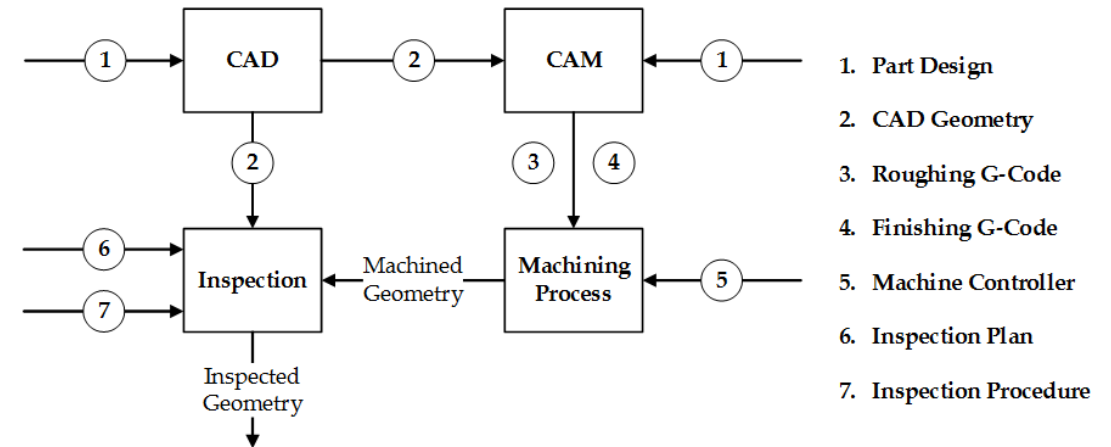
<https://ars-els-cdn-com.ezproxy.lib.vt.edu/content/image/1-s2.0-S0921509315301957-gr6.jpg>

Product Oriented C2P Attacks – Design and Designation

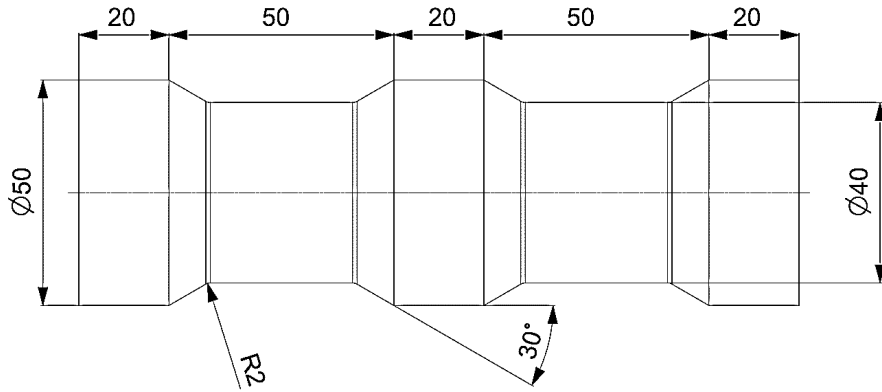
Attack Design and Designation Scheme



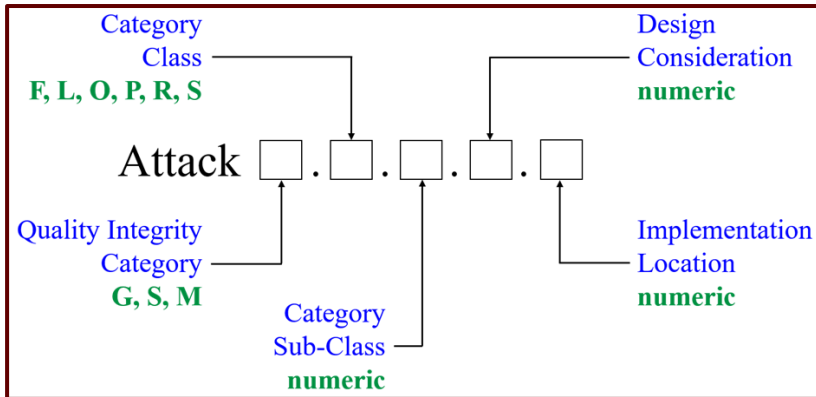
Consist of possible **system/product information** that can be used to design an attack with a lower likelihood of detection.



Product Oriented C2P Attack Examples



Simplified design of a spool



Attack Design and Designation Scheme

Product-oriented C2P attack examples

Attack Number		Geometric Change
A1	G.S.1-148-56	Decrease larger outside diameter 0.3 mm
A2	G.S.1-1568-235	Decrease larger outside diameter by 0.3 mm
A3	G.O.1-13569-2	Increase recess angle by 5°
A4	G.L.1-147-346	Dislocate the middle section by 2 mm to the right
A5	G.L.1-1478-346	Dislocate the middle section by 2 mm to the right
A6	G.F.1-15-34	Change the shape form of the two $\phi 40.0$ mm recess sections from a straight cylinder to a barrel shape

Reference. Shafae, M. S., Wells, L. J., & Purdy, G. T. (2019). Defending against product-oriented cyber-physical attacks on machining systems. *The International Journal of Advanced Manufacturing Technology*, 105, 3829-3850. <https://doi.org/10.1007/s00170-019-03805-z>

Product Oriented C2P Attack Examples

Attack Design & Designation

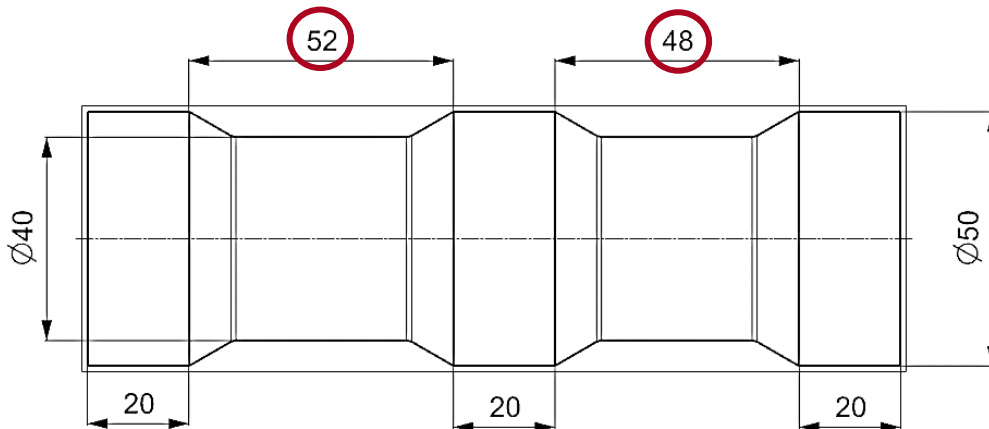
Attack G.L.1.147.346

- Change the position of the middle section (G.L.1)
- Design Considerations (147)
 1. Relatively small change hard to be recognized by operator
 4. Coordinated to attack only parts that will not be inspected
 7. Traditional ICS measures such as completion time and average machine power consumption
- Change the G-code (34) and access inspection plan (6)

Physical Impact

Impact on Process-Physics:

- Decreases the theoretical cutting time of roughing the first recess from 24.00 seconds to 22.80 seconds
- Increases the theoretical cutting time of roughing the second recess from 24.60 seconds to 25.80 seconds.

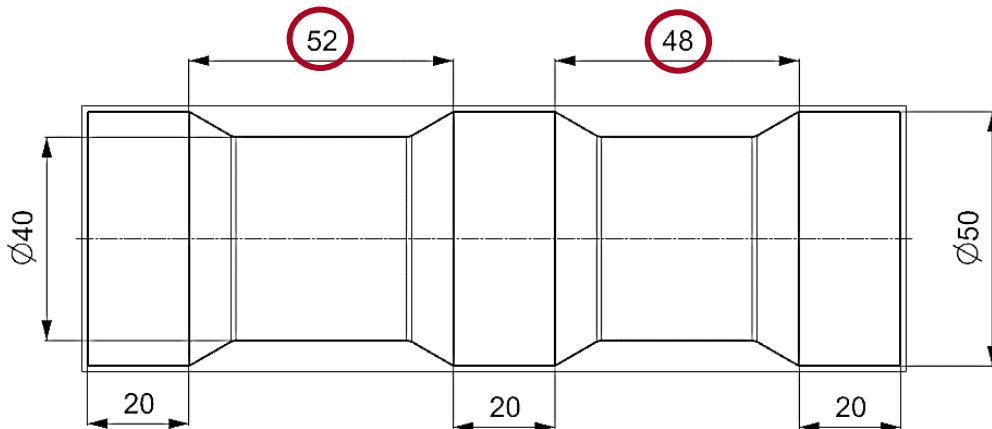


Product Oriented C2P Attack Examples

Attack Design & Designation

Attack G.L.1.1478.346

- **Change the position of the middle section (G.L.1)**
- **Design Considerations (1478)**
 1. Relatively small change hard to be recognized by operator
 4. Coordinated to attack only parts that will not be inspected
 7. Traditional ICS measures such as completion time and average machine power consumption
 8. **Compensate for the change in the recesses cutting time**
- **Change the G-code (34) and access inspection plan (6)**



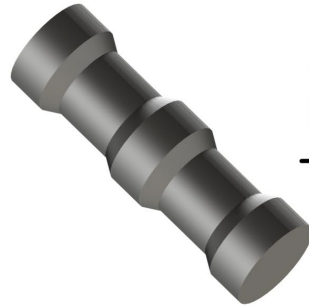
Physical Impact

Impact on Process-Physics:

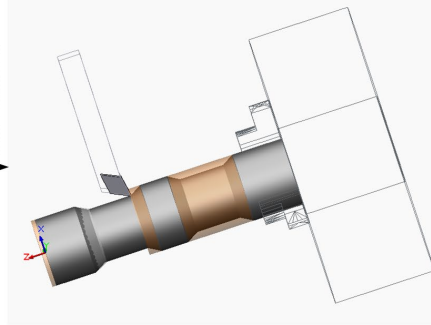
- Changes in the recesses cutting time will be mitigated by varying the feed rate.
- **Increases** the cutting forces needed to cut the **first recess**.
- **Decreases** the cutting forces needed to cut the **second recesses**.
- Average machine power consumption remains constant

Security Aware Inspection

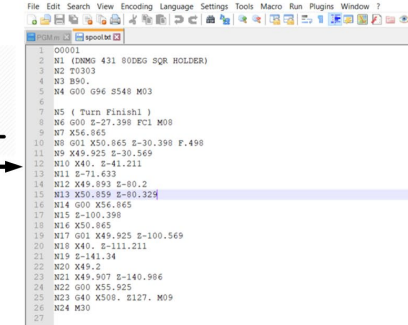
CAD File



CAM Software



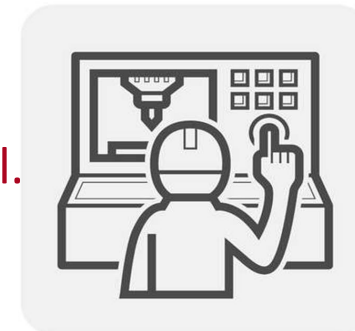
Tool Path File



Belikovetsky
et al. (2017)

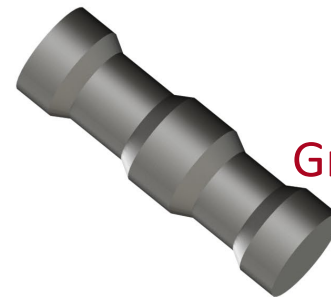
Sturm et al.
(2017)

Wells et al.
(2014)

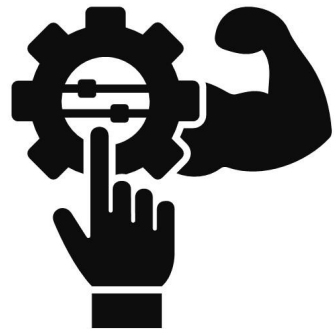


CNC Machine

Graves et al.
(2019)

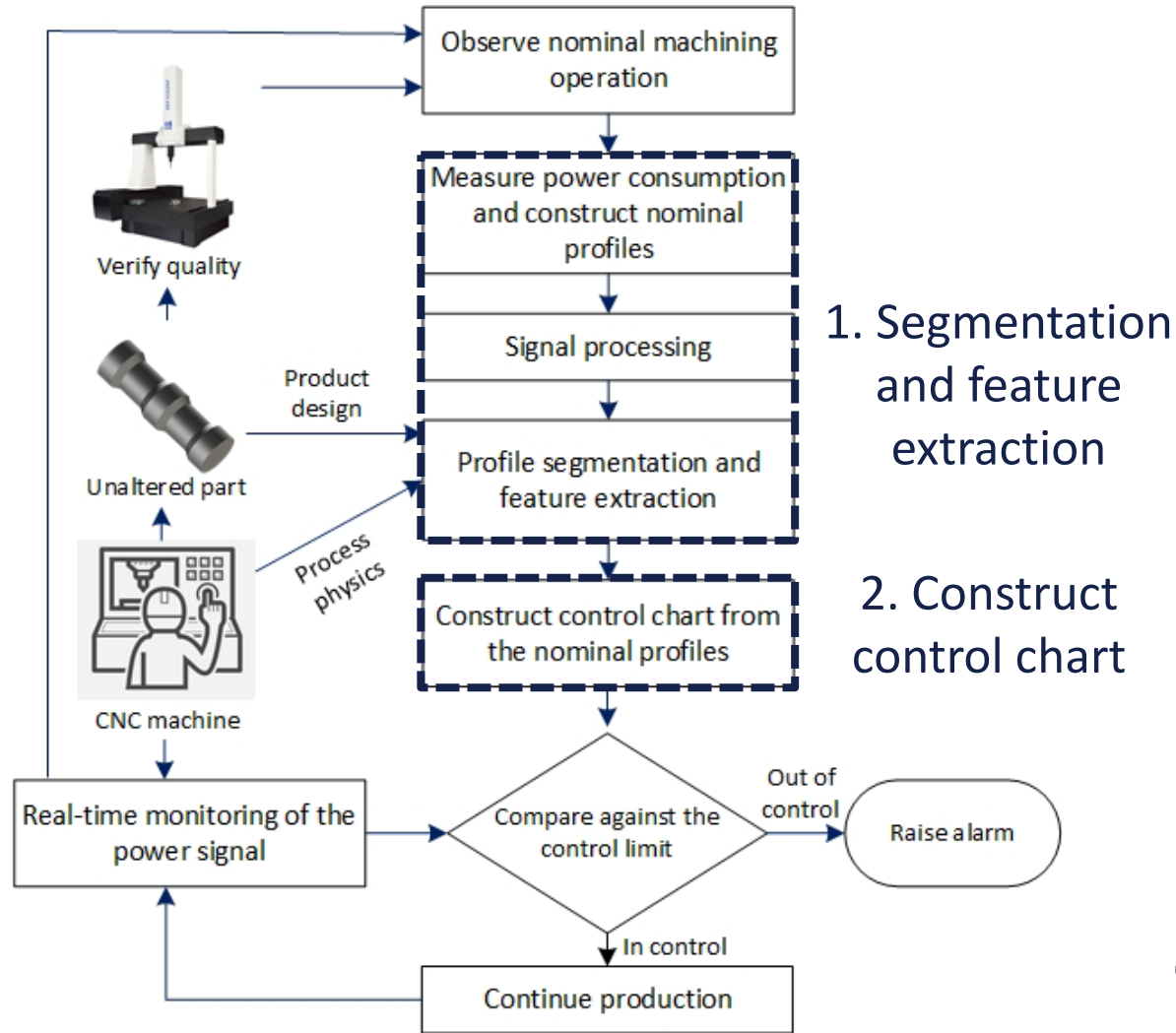


Altered Part

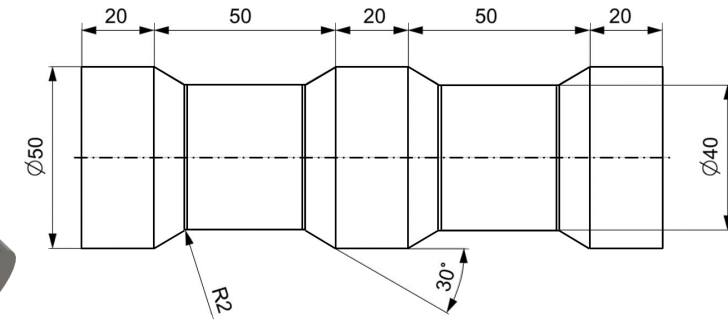


Inspection

Physics-Informed Attack Detection

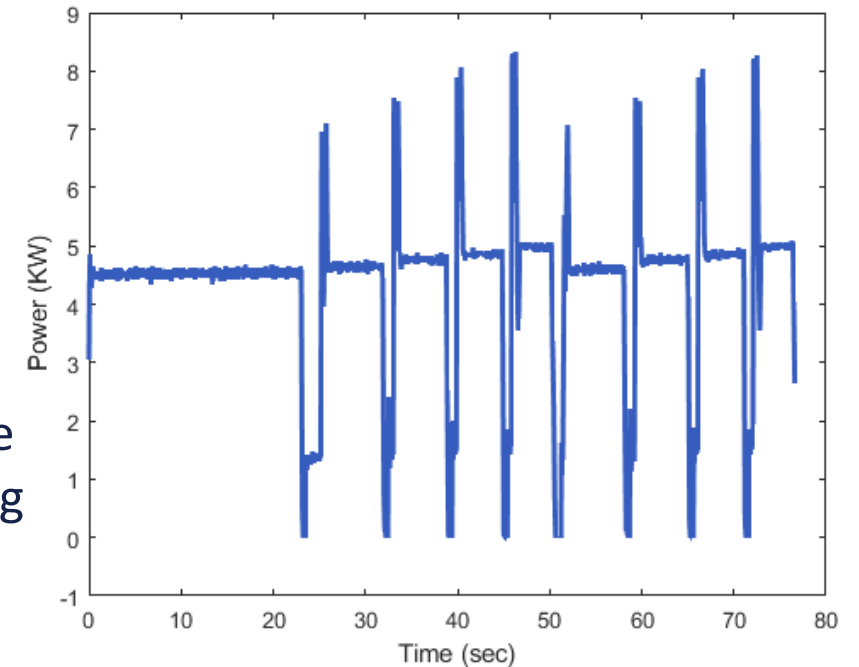


Proposed attack detection algorithm

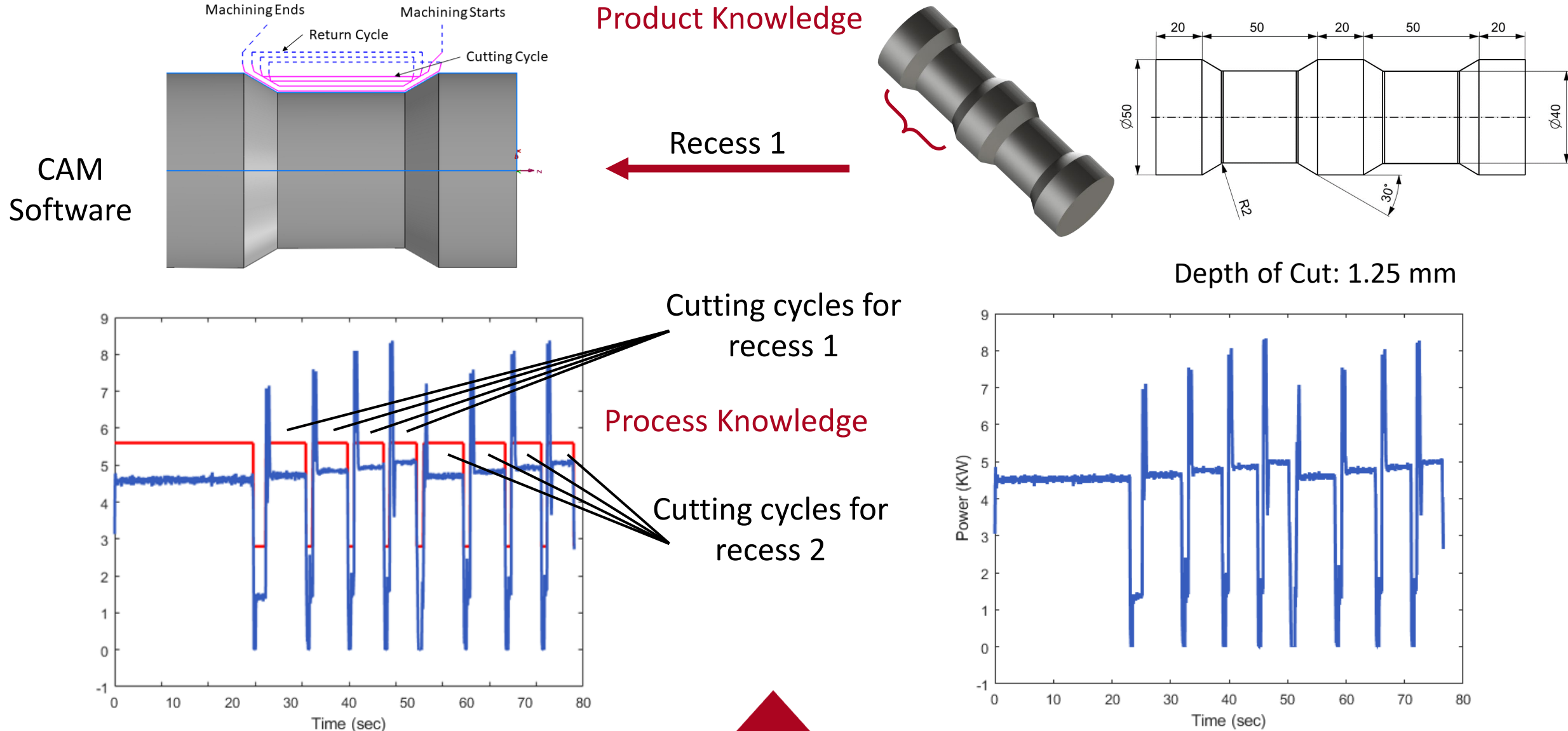


Depth of Cut: 1.25 mm

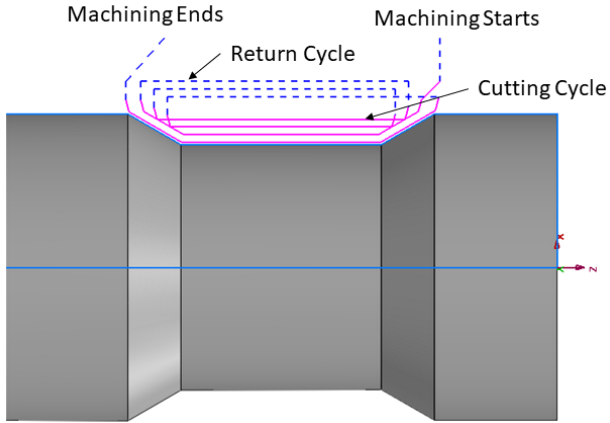
Power profile
during turning
operation



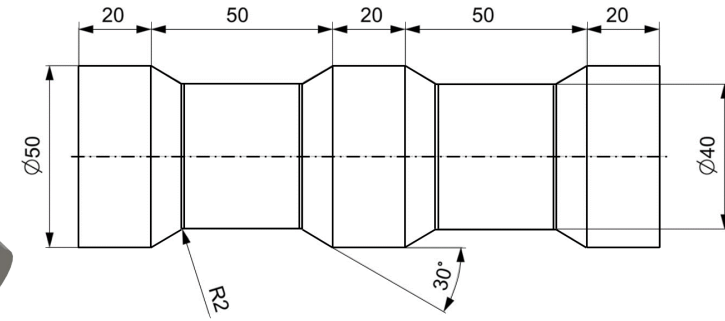
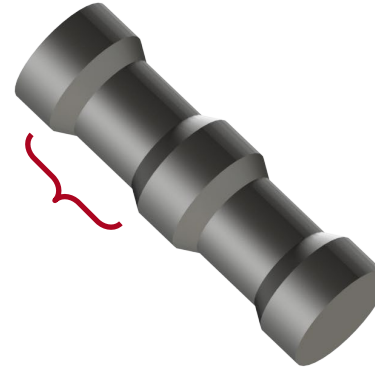
Physics-Informed Attack Detection



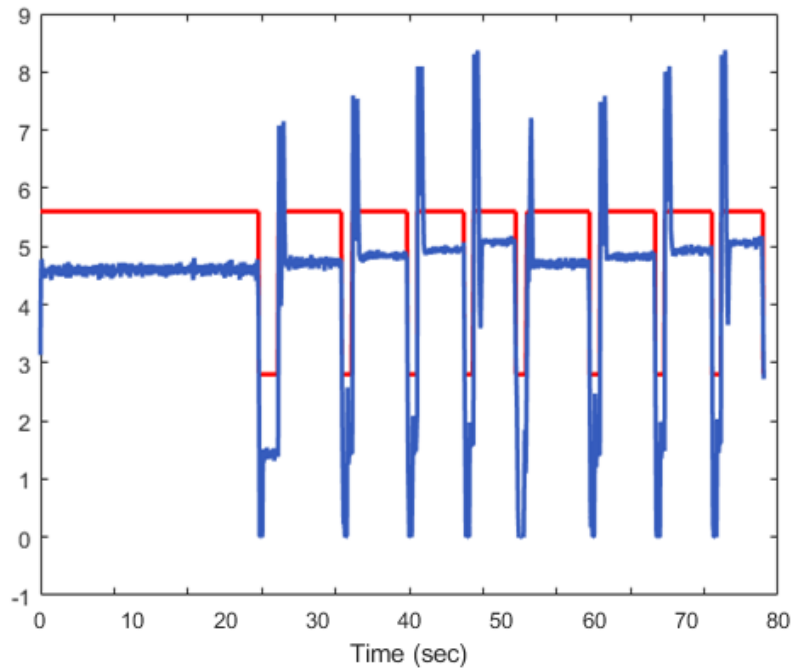
Feature Extraction



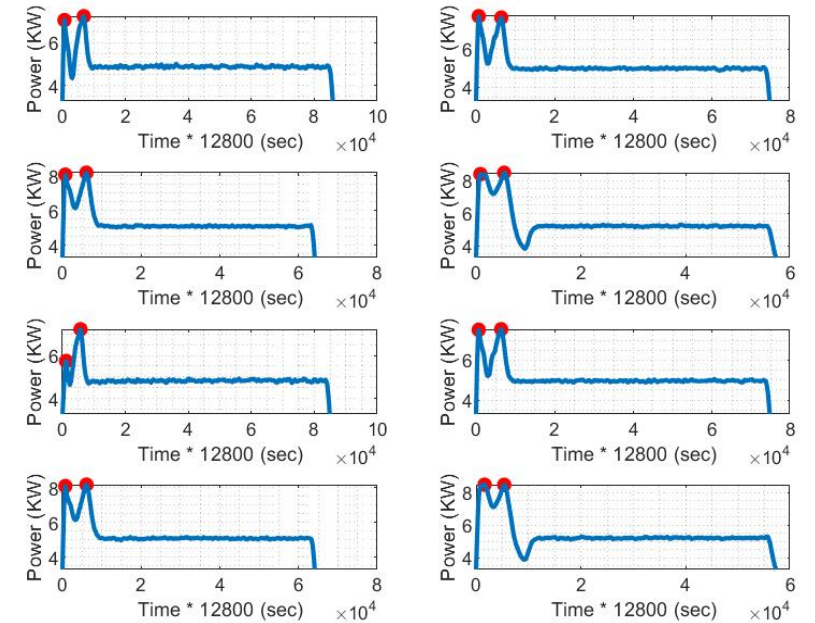
Recess 1



Depth of Cut: 1.25 mm



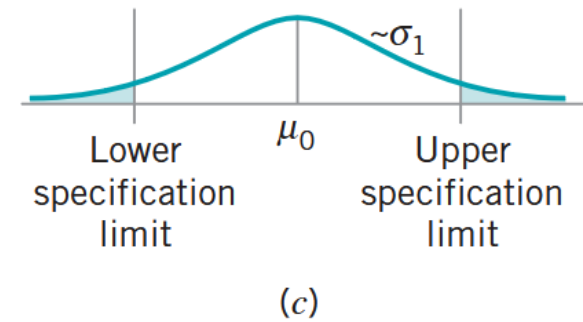
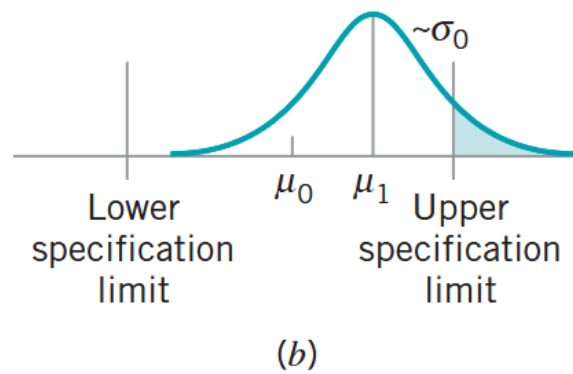
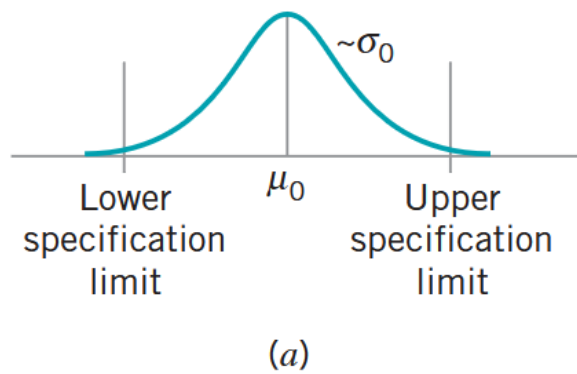
Profile Segments



Control Chart for Variables

Objective. Monitoring the **mean** and **variability**

- a) Mean and standard deviation at nominal levels
- b) Shift in the process mean ($\mu_1 > \mu_0$)
- c) Shift in process standard deviation ($\sigma_1 > \sigma_0$)



Control Chart – Single Variate

Definition. A statistical tool used to monitor the stability of a process based on a single quality characteristic.

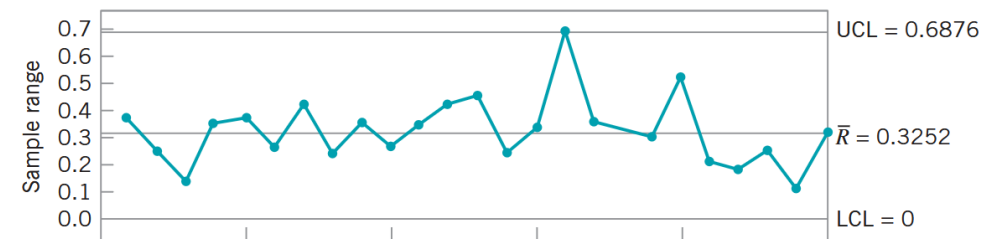
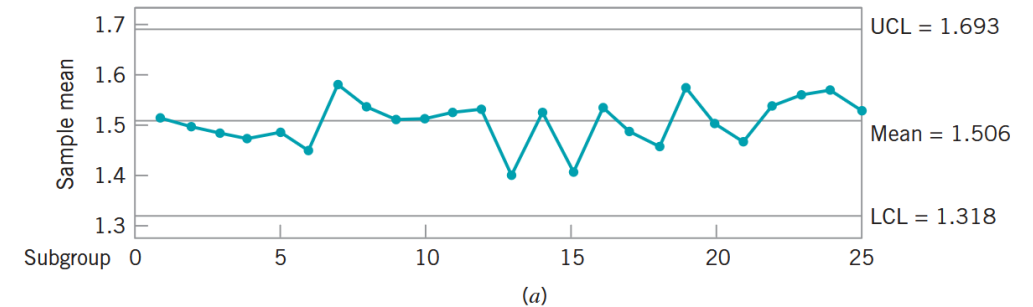
- Monitors one variable at a time.
- Useful for identifying trends, shifts, or outliers in a process.
- **Example.** Tracking the diameter of a machined part over time.
- Common types include $\bar{X}$ chart (mean), R chart (range), and S chart (standard deviation).

Control Limits for the $\bar{X}$ Chart

$$UCL = \bar{\bar{x}} + A_2 \bar{R}$$

$$\text{Center line} = \bar{\bar{x}}$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R}$$



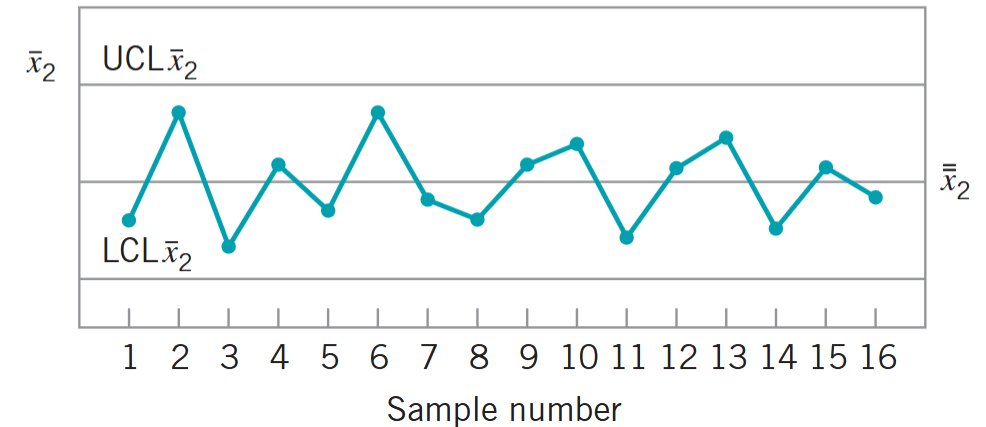
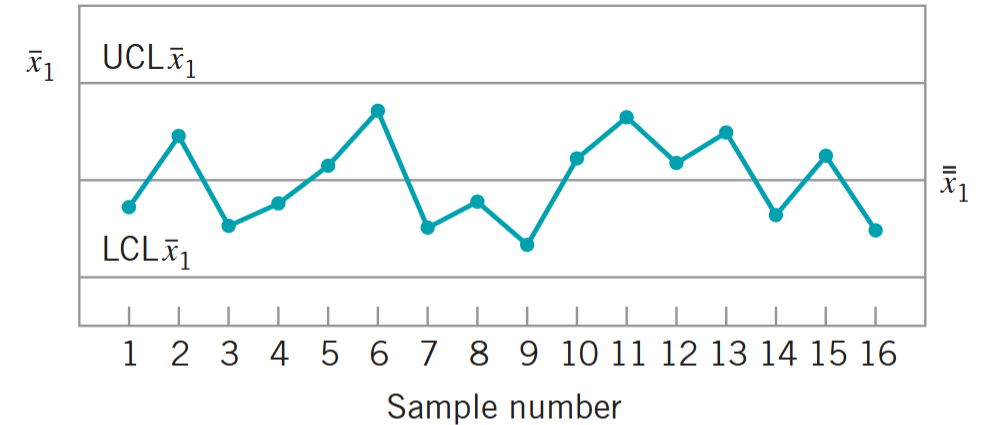
Control Limits for the R Chart

$$UCL = D_4 \bar{R}$$

$$\text{Center line} = \bar{R}$$

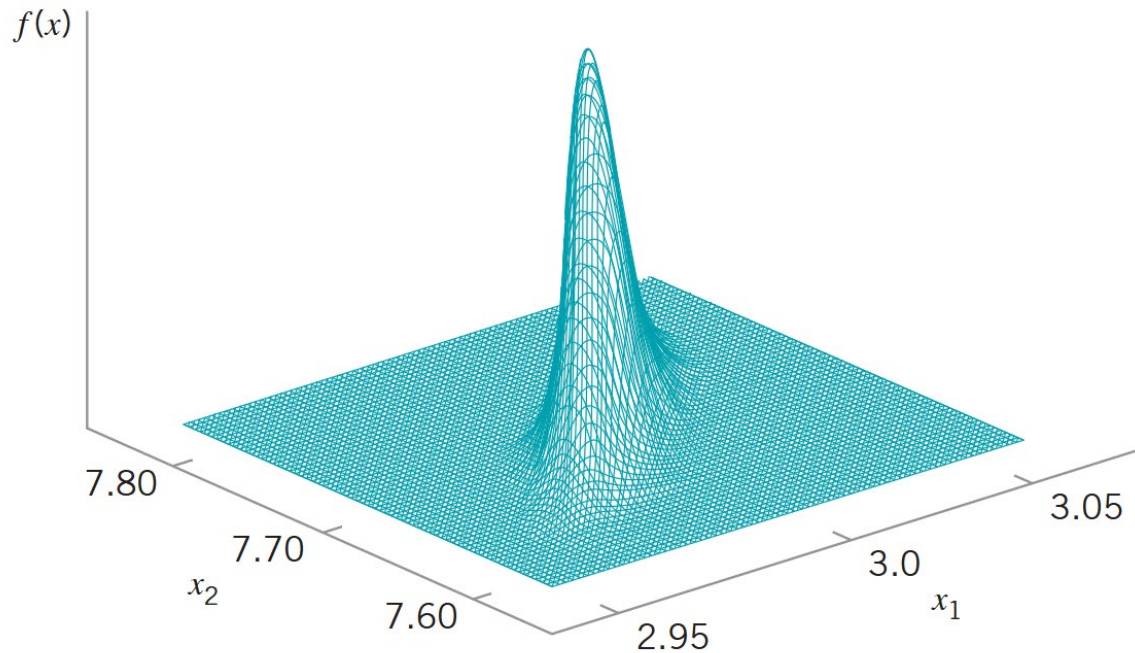
$$LCL = D_3 \bar{R}$$

Control Chart – Multi Variate

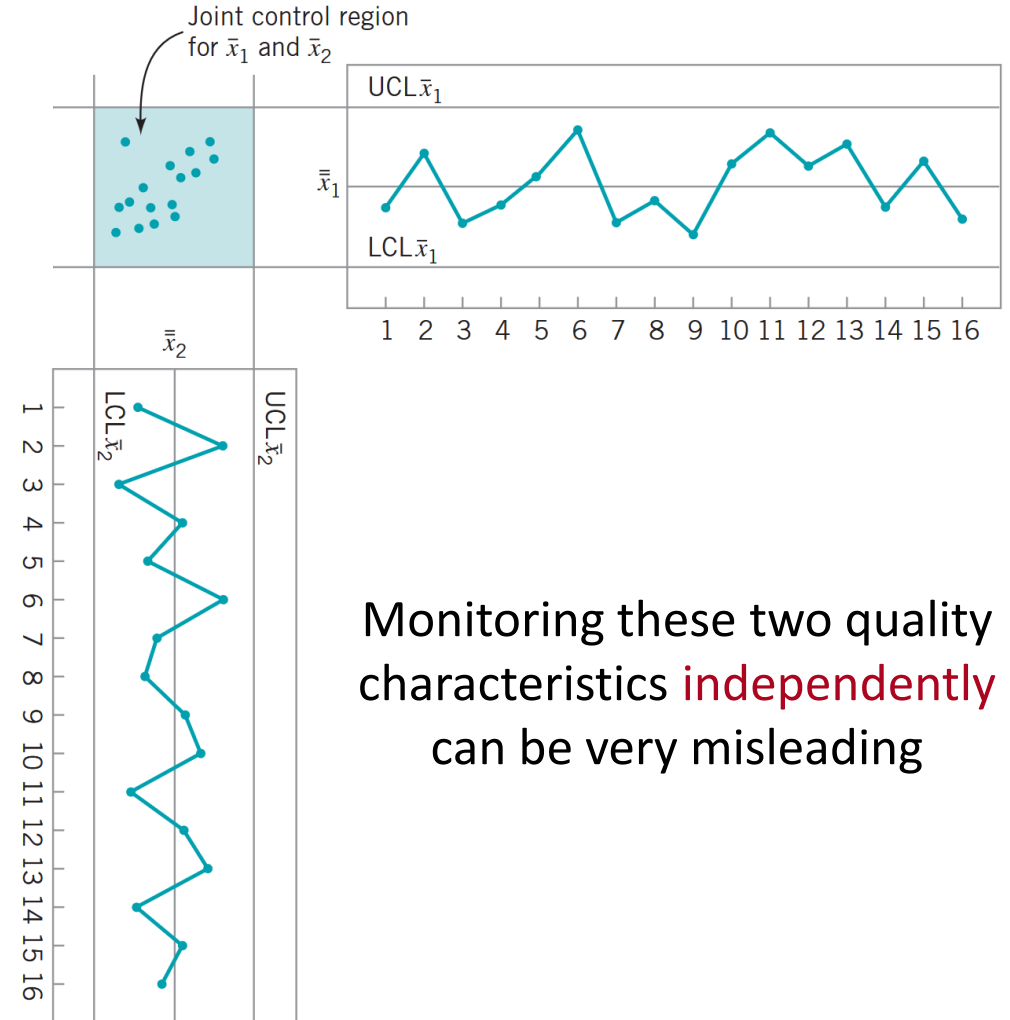


Control charts for inner (x_1) and outer (x_2) bearing diameters

Control Chart – Multi Variate



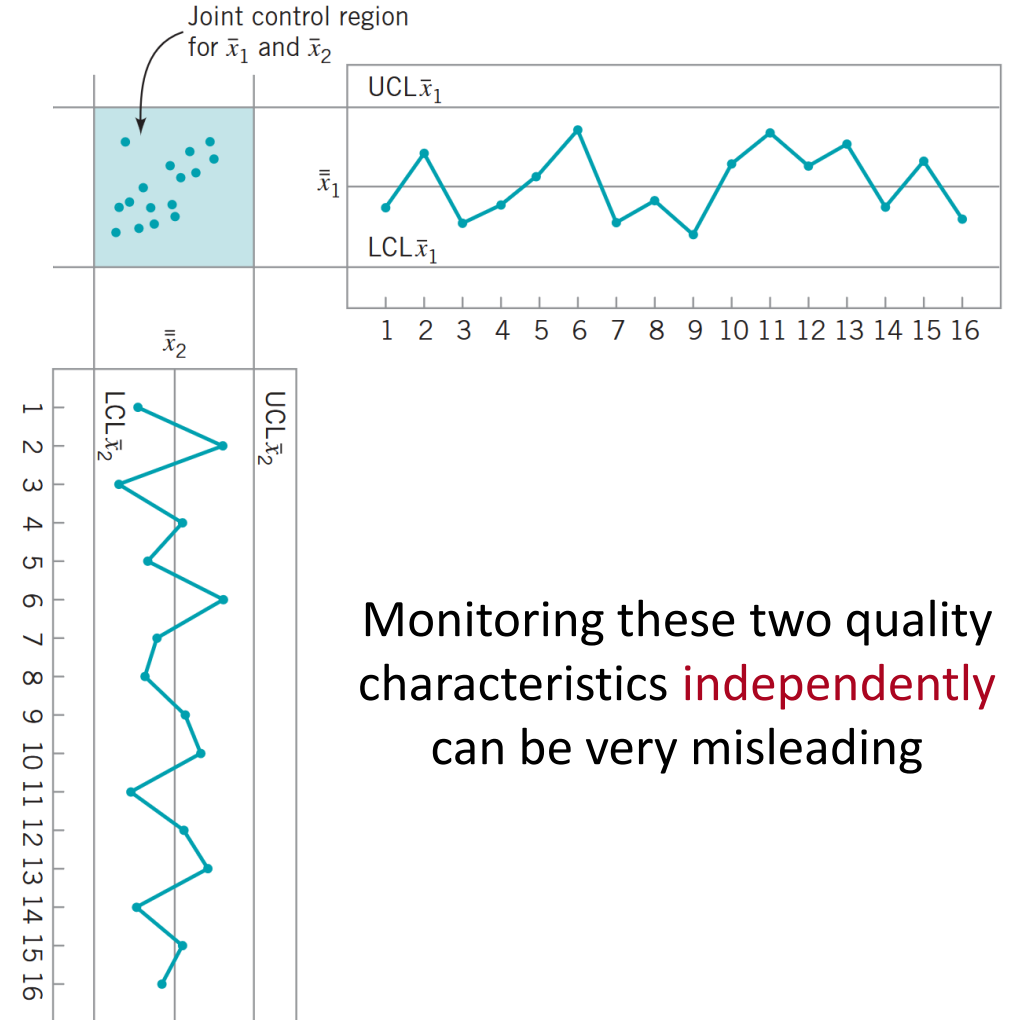
A multivariate normal distribution with two variables (bivariate normal)



Control Chart – Multi Variate

Monitors **multiple quality characteristics simultaneously**

- Analyzes correlation among variables.
- Detects multidimensional outliers and shifts in the process
- Common type: Hotelling T^2 control chart



Control Chart for Geometric Integrity

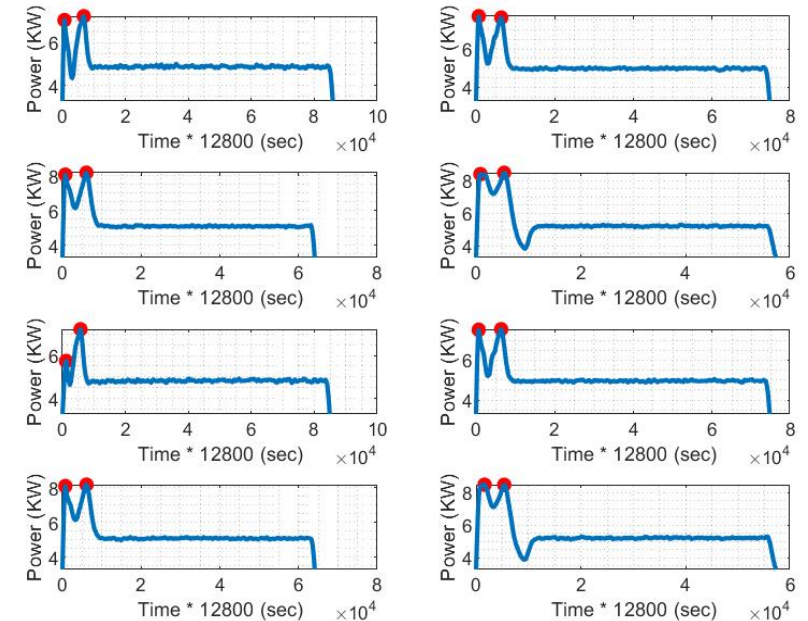
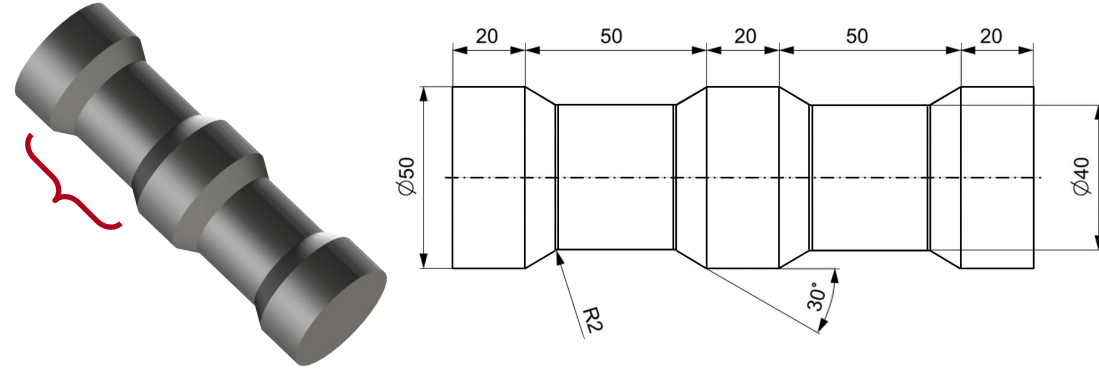
Machining time for recess 1 & 2

Hotelling T^2 control chart

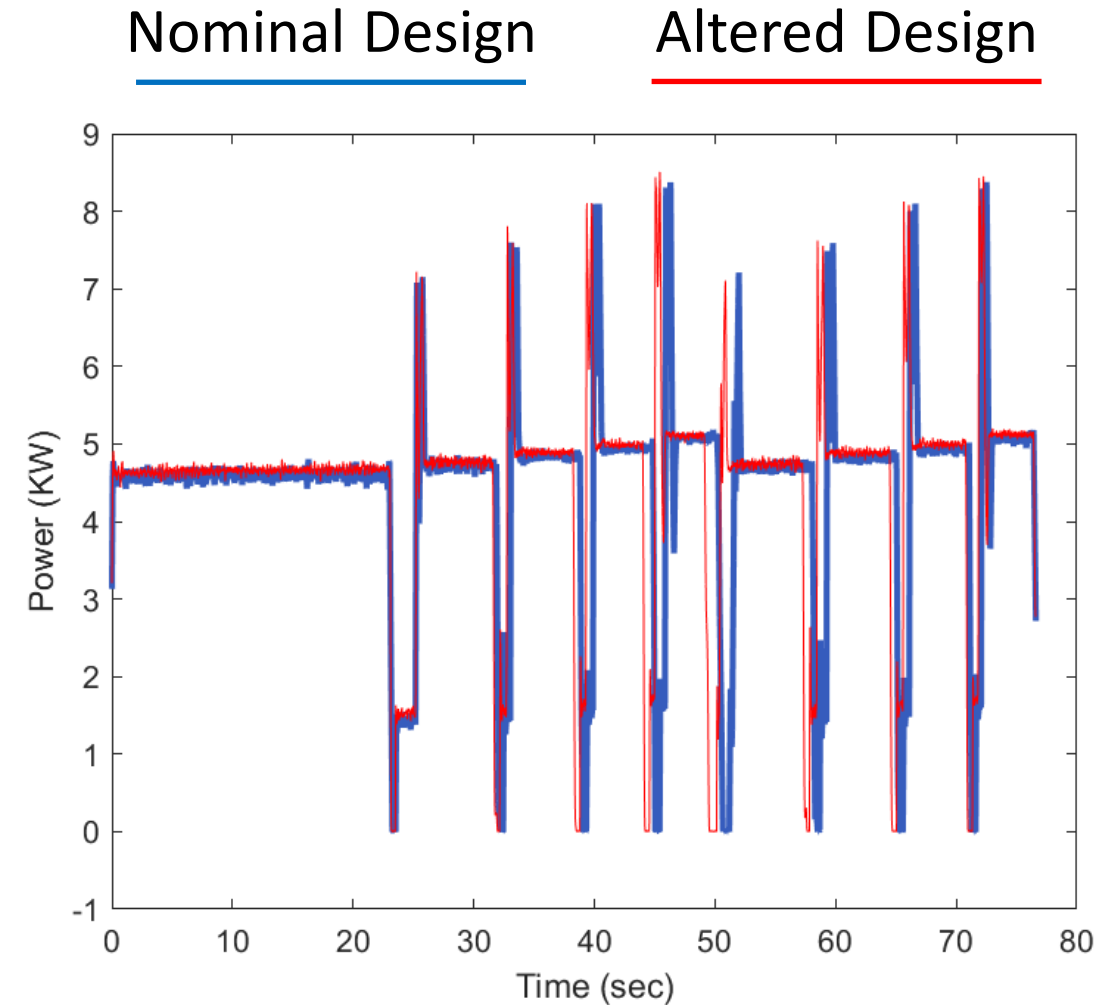
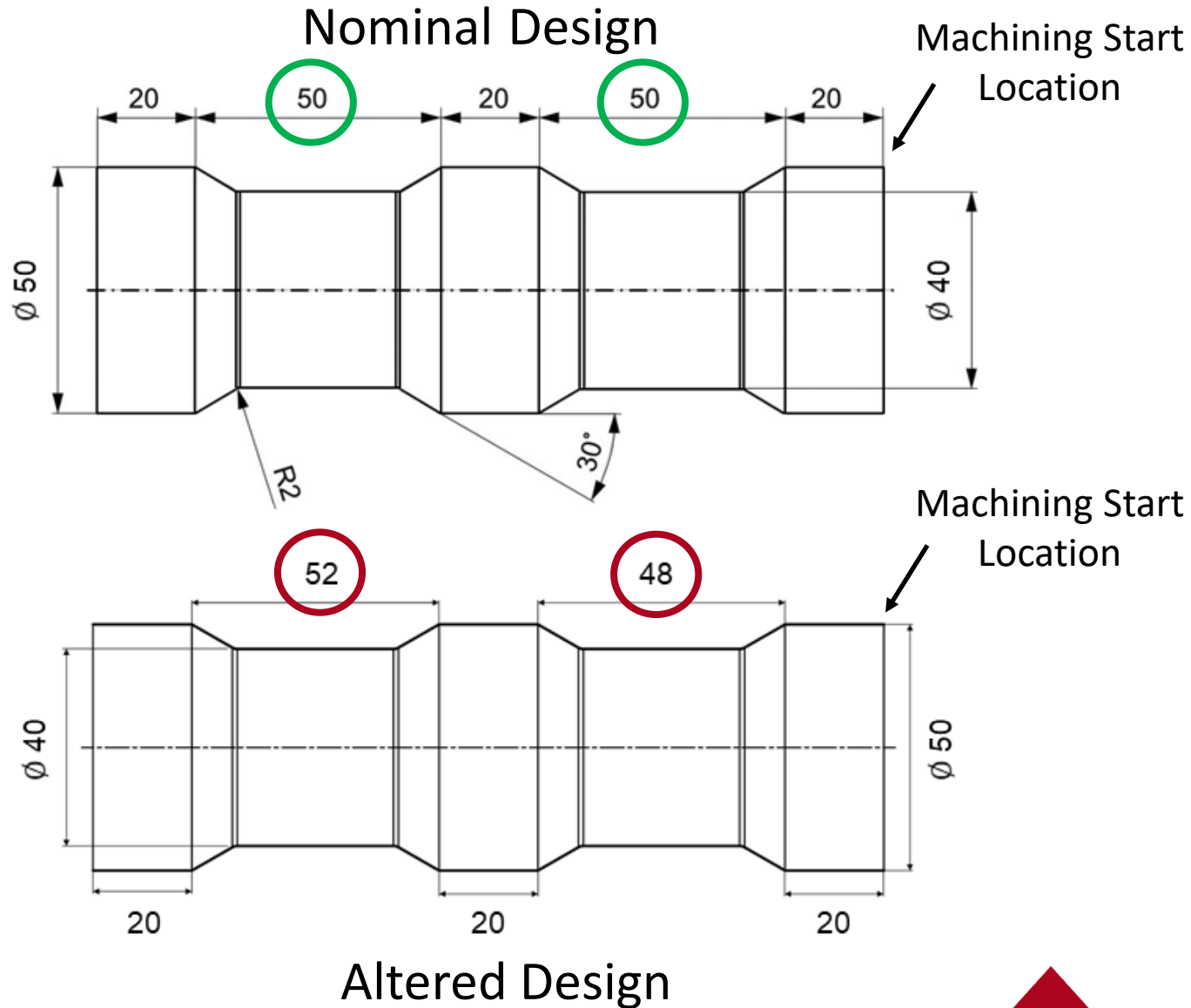
$$T^2 = (Y - \bar{Y})^T S^{-1} (Y - \bar{Y})$$

$$UCL = \chi^2_{\alpha, p}$$

$$LCL = \frac{p(m-1)(n-1)}{mn - m - p + 1} F_{1-\alpha, p, mn-m-p+1}$$

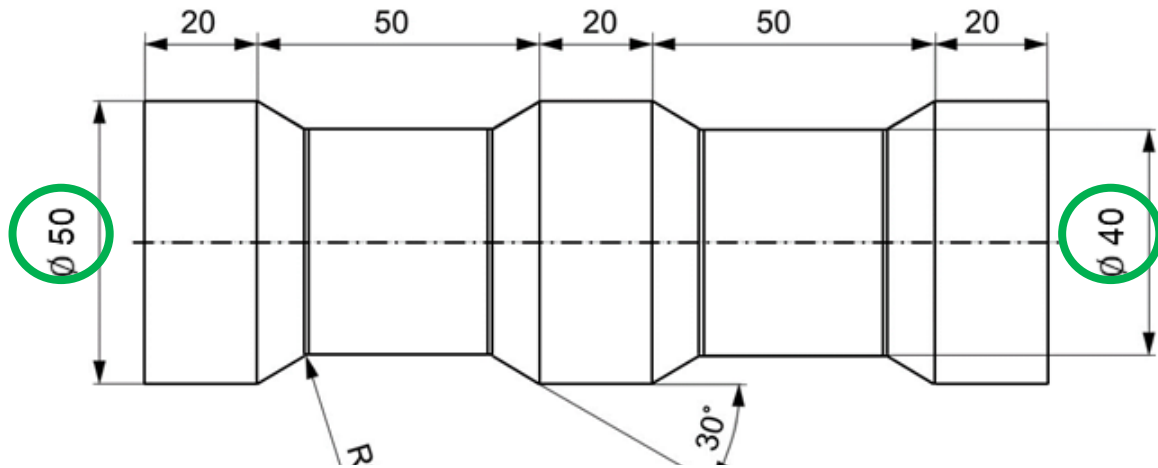


Implemented Attack and Result

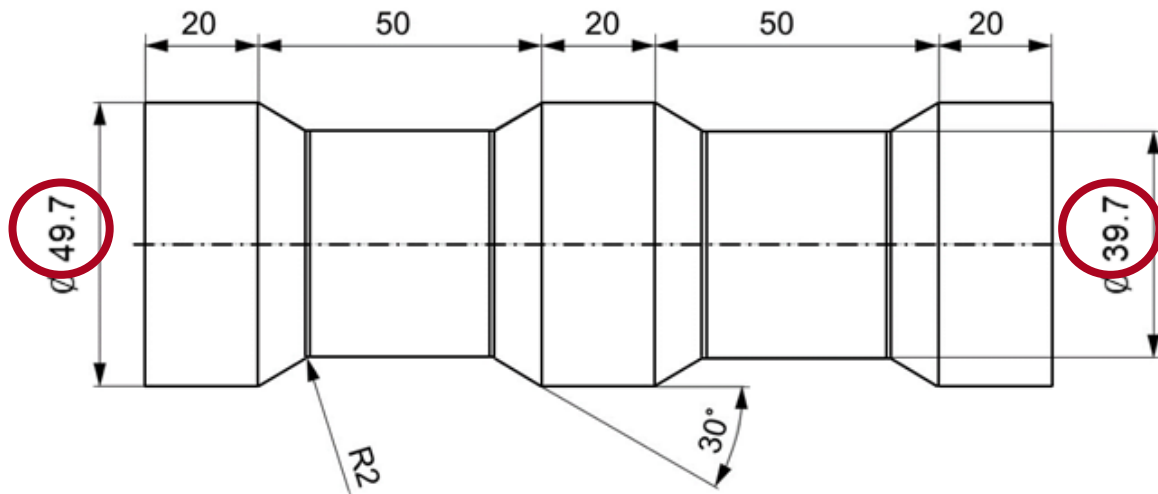


Implemented Attack and Result

Nominal Design



$$MRR = d(\text{depth of cut}) \times f(\text{feed rate}) \times v(\text{cutting speed})$$



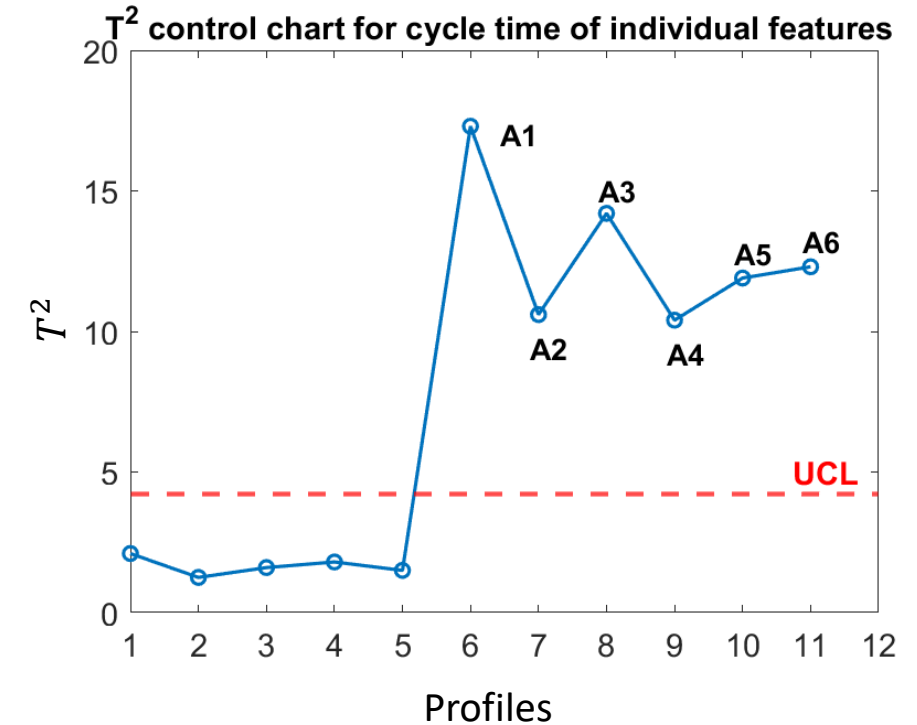
Altered Design

Decrease feed rate to mitigate the impact of **increased depth of cut**

Control Chart for Geometric Integrity

Attack Number		Geometric Change
A1	G.S.1.148.56	Decrease larger outside diameter 0.3 mm
A2	G.S.1.1568.235	Decrease larger outside diameter by 0.3 mm
A3	G.O.1.13569.2	Increase recess angle by 5°
A4	G.L.1.147.346	Dislocate the middle section by 2 mm to the right
A5	G.L.1.1478.346	Dislocate the middle section by 2 mm to the right
A6	G.F.1.15.34	Change the shape form of the two $\phi 40.0$ mm recess sections from a straight cylinder to a barrel shape

Product-oriented C2P attack examples



Attack detection using the proposed approach

Significance of Results

- Novel **security-aware spindle power monitoring** approach for detecting attacks on **dimensional and geometric integrity** of products
- Provides foundation for the following contributions:
 - Extracting granular **temporal** and **amplitude** features
 - **Diagnosis** on the changes made by attacks
 - Developing metrics to distinguish **attacks** from **inadvertent mistakes**

